

Modeling infectious diseases using R

Practical Session:

Static and Dynamic aspects of SIR model

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First Training workshop, Cape Town, South Africa
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Sustainable Training in Epidemiological Analysis and Modelling
for Data-Driven Decision Making in Africa

What do we cover in this practical session ?

- Simple transmission models in R.
- SIR: static aspect (time homogeneity).
- SIR: dynamic aspects.
- Vaccination.
- Software:
 - R.
 - The **deSolve** package in R.

R requirements

- Basic knowledge in R.
- Programming of a user defined simple function.
- Why R ?
 1. Free.
 2. Fast (not in our case).
 3. Updated.
 4. Documented.

R programs for the practical session

- HTML file for slides 5-78:
 - [eR_prog1_STEAM3D_2026_V1.html](#)
- Two R programs for slides 5-78:
 - R program:
 - [ModelingIDinR1_V1_Stat&Dynam_10_2026.R.](#)
 - R requirements: basic R editor.
 - R markdown:
 - [eR_prog1_STEAM3D_2026_V1.Rmd.](#)
 - R requirements: R Studio.

Part 1:
Time homogeneity (the static model)

Interactive tutorial: the HTML file

The screenshot shows a web browser window with the address bar displaying the file path: `C:/Ziv_Temp_2023/Tanylee_STEAM-3D/Workshop1_12_10_26/ShortCourse/Practicalessions_New/er_prog1_STEAM3D_2026-V1.html`. The browser has tabs for "er_prog1_STEAM3D_2026-V1.html" and "Open in Browser". A search bar is visible in the top right corner.

On the left side of the page, there is a sidebar with a list of sections:

- 1. The SIR model
- 2. The SIR model in a closed population
- 3. The SIR model in open population
- 4. Threshold and Equilibrium Values

The main content area displays the date "10-09-2026" and the author "Ziv Shkedy and Rashider Aloni" based on Chapter 2-4 in the book *Modeling infectious diseases parameters based on serological and social contact data* (<https://www.openintro.org/book/biostat/>).

Modeling infectious diseases using R - the SIR model

1. The SIR model

The starting point in this chapter is the SIR model in endemic equilibrium. In particular, we assume that all parameters are constant over time and the focus goes to the **age** of the individual as the time scale of primary interest. In Section XXX equilibrium values for the SIR-model with vital dynamics and a constant population were derived in case of homogeneous mixing. In practice however age is a predominant factor in the way an infection spreads through a population given that children have had fewer years of exposure than adults.

Formulation of the SIR model

Extending the SIR model in terms of changes over **age** and **time** yields the following system of partial differential equations:

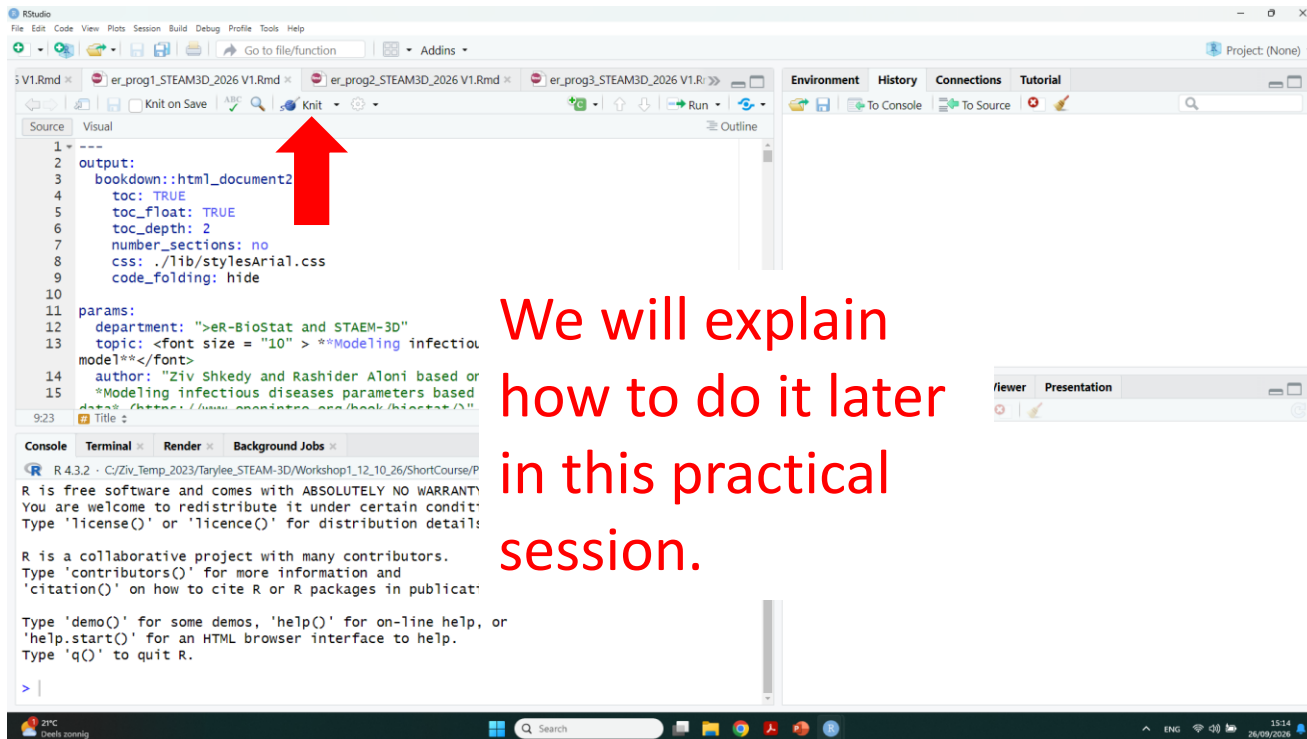
$$\frac{\partial S(a, t)}{\partial t} + \frac{\partial S(a, t)}{\partial a} = -\mu(a)S(a, t) - \beta(a)S(a, t)I(a, t) + \gamma(a)I(a, t) - \mu(a)S(a, t)$$

The browser's taskbar at the bottom shows the system clock as 15:10 on 26/09/2026, along with various system icons and a search bar.

Section 1 & 2 in the HTML file.

How to produce the HTML file ?

- Open R Studio.
- Open the file: eR_prog1_STEAM3D_2026_V1.Rmd.
- Knit the file.



SIR model :time homogeneity

$$\frac{dS(a)}{da} = -\lambda S(a)$$

$$\frac{dI(a)}{da} = \lambda S(a) - \sigma I(a)$$

$$\frac{dR(a)}{da} = \sigma I(a)$$



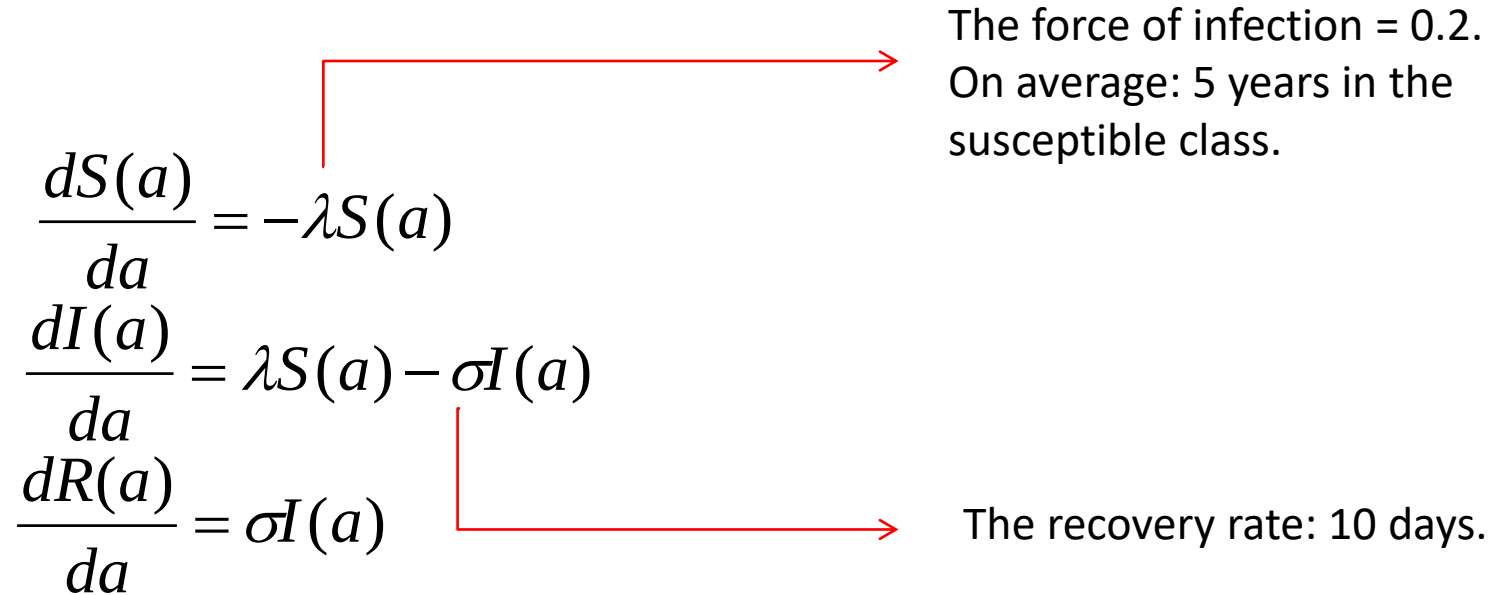
The force of infection: the rate in which individuals are infected



The recovery rate: the rate in which individuals recovered and move to the immune class.

Assumption: life long immunity.

SIR model: time homogeneity



The diagram shows three differential equations for the SIR model. Red arrows point from the parameters in the equations to their interpretations:

- From λ in $\frac{dS(a)}{da} = -\lambda S(a)$ to: The force of infection = 0.2. On average: 5 years in the susceptible class.
- From σ in $\frac{dR(a)}{da} = \sigma I(a)$ to: The recovery rate: 10 days.

$$\frac{dS(a)}{da} = -\lambda S(a)$$
$$\frac{dI(a)}{da} = \lambda S(a) - \sigma I(a)$$
$$\frac{dR(a)}{da} = \sigma I(a)$$

The unit of the parameters are in years

Transmission models in R

- We need to integrate the system of the ordinary differential equation.
- deSolve package in R.
- Numerical integration using of ODE system.

$$\begin{aligned}\frac{dS(a)}{da} &= -\lambda S(a) \\ \frac{dI(a)}{da} &= \lambda S(a) - \sigma I(a) \\ \frac{dR(a)}{da} &= \sigma I(a)\end{aligned}$$

Transmission models in R

SIR model

$$\begin{aligned}\frac{dS(a)}{da} &= -\lambda S(a) \\ \frac{dI(a)}{da} &= \lambda S(a) - \sigma I(a) \\ \frac{dR(a)}{da} &= \sigma I(a)\end{aligned}$$

Specification in R

- Model parameters.
- State variables (the value of the parameters at age (time) zero).
- Time range (=age range) for integration.

Transmission models in R

R program

Model parameters



λ, σ

State variables



$S(0), I(0), R(0)$

Time range for
integration



Integrate from age zero
to age 40

Specification of the
model



$$\begin{aligned}\frac{dS(a)}{da} &= -\lambda S(a) \\ \frac{dI(a)}{da} &= \lambda S(a) - \sigma I(a) \\ \frac{dR(a)}{da} &= \sigma I(a)\end{aligned}$$

Specification of the model parameters in R

```
> parameters <- c(lambda = 0.2, v=36.5)
> parameters
lambda      v
  0.2     36.5
```

The force of
infection = 0.2
years.

$$5^{-1} = 0.2$$

Recovery rate 10
days.

$$\left(\frac{10}{365}\right)^{-1} = 36.5$$

The unit of the
parameters are
in years⁻¹

The state variables (initial values at age 0)

- Let us assume that the cohort size is 5000.
- At age=0:

$$S(0) = 4999$$

$$I(0) = 1$$

$$R(0) = 0$$

- Specification in R:

```
> state <-  
c(X=4999, Y=1, Z=0)  
> state  
      X      Y      Z  
4999    1     0
```

Specification of the model in R

$$\left. \begin{aligned} \frac{dS(a)}{da} &= -\lambda S(a) \\ \frac{dI(a)}{da} &= \lambda S(a) - \sigma I(a) \\ \frac{dR(a)}{da} &= \sigma I(a) \end{aligned} \right\}$$

```
SIR<-  
function(t,state,parameters)  
{  
  with(as.list(c(state,  
parameters))),  
  {  
    dX <- -lambda*X  
    dY <- lambda*X - v*Y  
    dZ <- v*Y  
    list(c(dX, dY, dZ))  
  })  
}
```

We ask from the function to
return the values of S, I and R ¹⁵

Specification of the time units for the integration

$$\begin{aligned}\frac{dS(a)}{da} &= -\lambda S(a) \\ \frac{dI(a)}{da} &= \lambda S(a) - \sigma I(a) \\ \frac{dR(a)}{da} &= \sigma I(a)\end{aligned}$$

- The solution of the model: numerical integration.
- Time units: age.
- Integration from age 0 to age 40 by unit of 0.01 years

```
> times<-seq(0,40,by=0.01)
> times
[1] 0.00 0.01 0.02 0.03 0.04 0.05 0.06 0.07 0.08 0.09 0.10 0.11
[13] 0.12 0.13 0.14 0.15 0.16 0.17 0.18 0.19 0.20 0.21 0.22 0.23
[25] 0.24 0.25 0.26 0.27 0.28 0.29 0.30 0.31 0.32 0.33 0.34 0.35
[37] 0.36 0.37
```

Running the model

```
require(deSolve)  
out<-as.data.frame(ode(y=state,times=times,func=SIR,parms=parameters))
```

The state variables:
the values at age 0.

The model
parameters: force of
infection (0.2) and
recovery rate (10
days)

Running the model

```
require(deSolve)  
out<-as.data.frame(ode(y=state,times=times,func=SIR,parms=parameters))
```



Time range for
integration.



The model



Numerical integration
using ordinary
differential equation

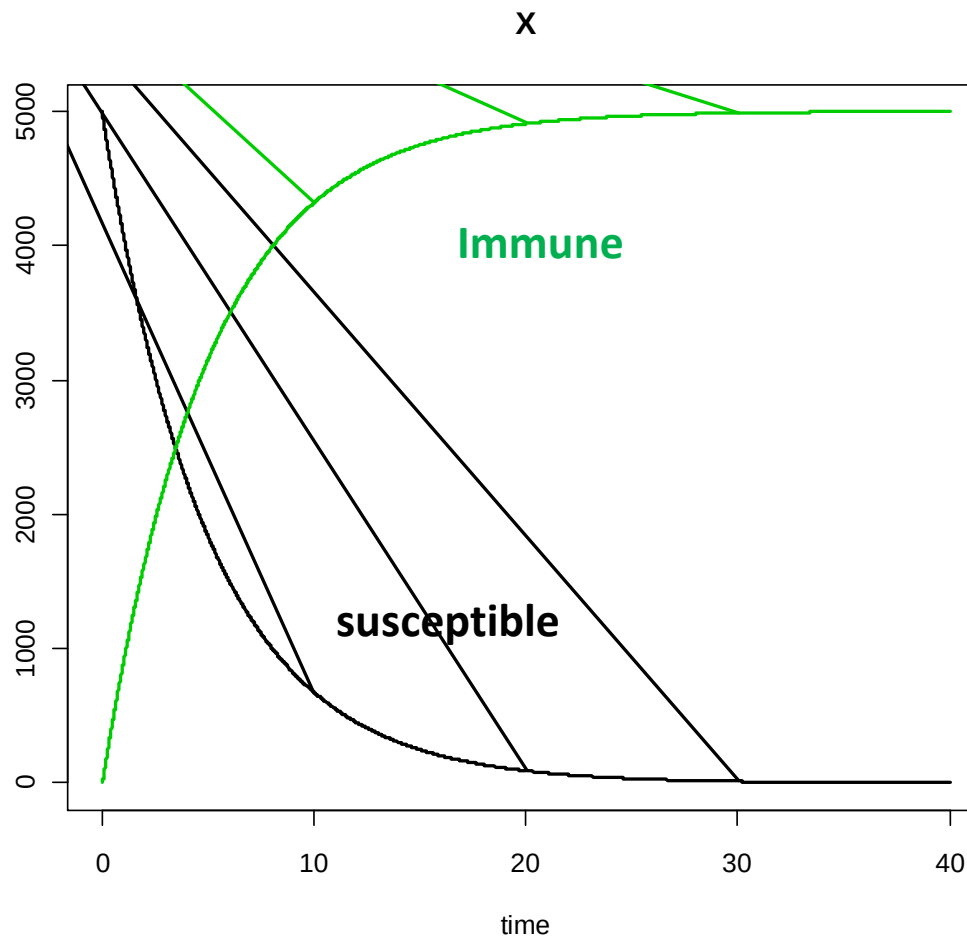
Solution

```
> require(deSolve)
> out <-
as.data.frame(ode(y=state, times=times, func=SIR, parms=
parameters))
> head(out)
```

	time	X	Y	Z
1	0.00	4999.000	1.000000	0.000000
2	0.01	4989.012	9.061818	1.926190
3	0.02	4979.044	14.641580	6.314481
4	0.03	4969.096	18.498345	12.405853
5	0.04	4959.168	21.159066	19.673391
6	0.05	4949.259	22.989501	27.751380

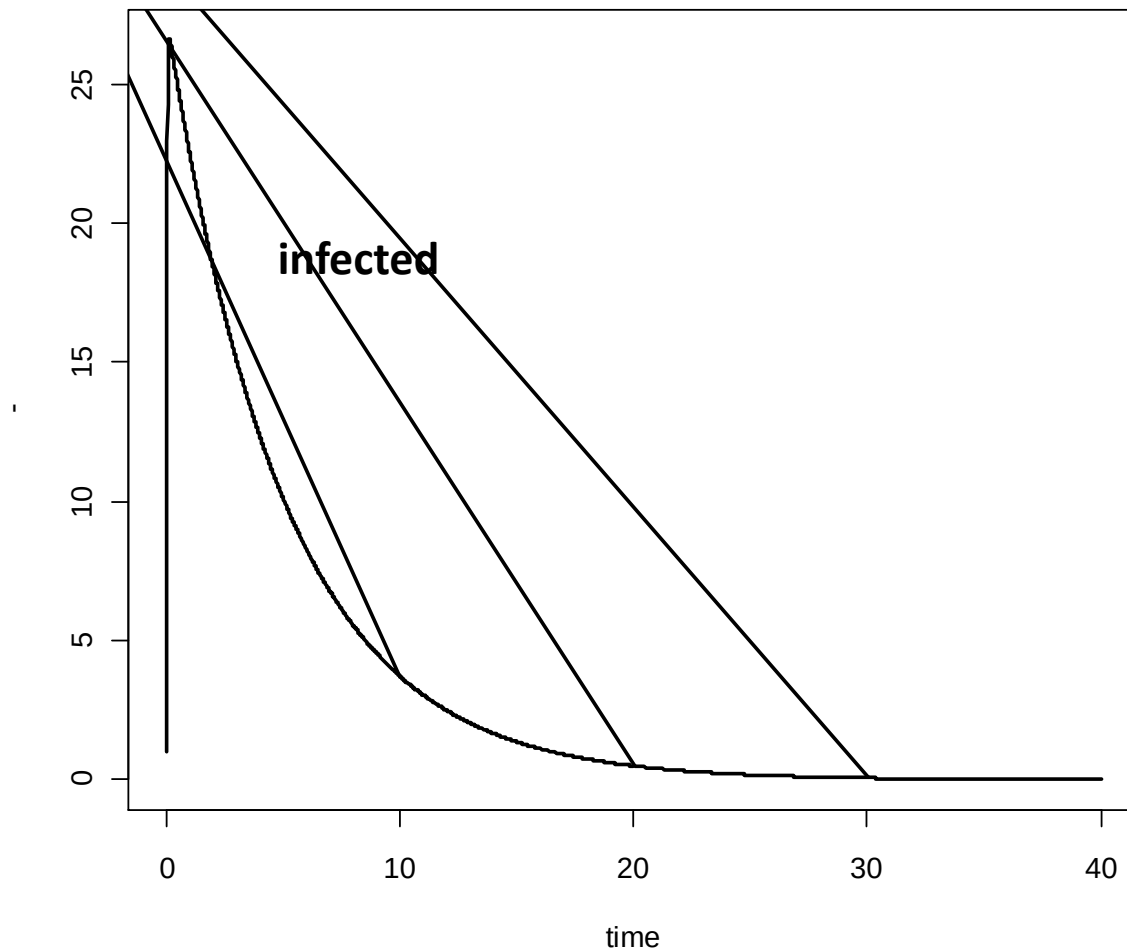
Graphical output

```
> plot (times,out$X ,type="l",main="X", xlab="time", ylab="-",lwd=2)  
> lines(times,out$Z,col=3,lwd=2)
```



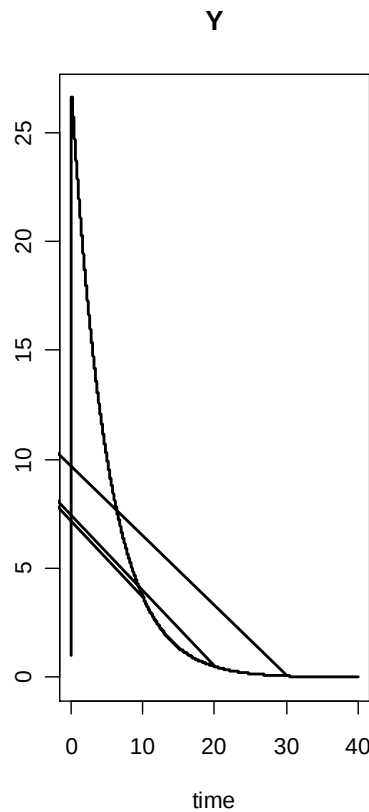
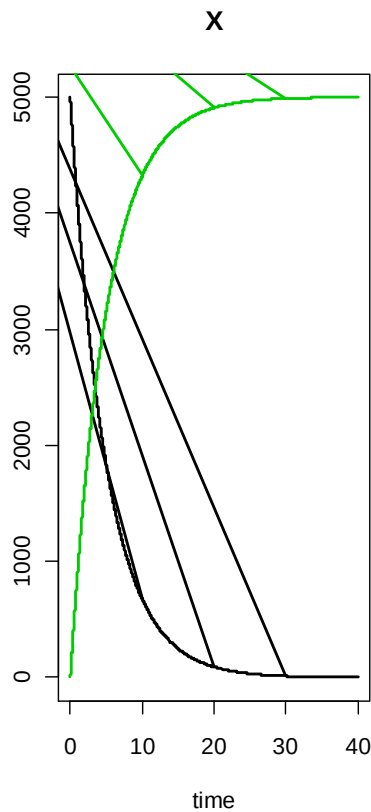
Graphical output

```
> plot (times,out$Y ,type="l",main="Y", xlab="time", ylab="-",lwd=2)
```



Graphical output

SIR model



What do we see here ?

Number of infected individuals at each age.

Slides 13-23: example 1 in the HTML

1. The SIR model
2. The SIR model in a closed population
Example 1 ($\lambda = 0.2, D=10$ days)
Example 2 ($\lambda = 0.2, D=60$ days)
Example 3: age dependent force of infection
3. The SIR model in open population
4. Threshold and Equilibrium Values

Example 1 ($\lambda = 0.2, D=10$ days)

SIR model with $\lambda = 0.2$ and $D = 10$ days ($\nu = 36.5$)

For a population in which the average age at infection is 5 years (i.e., the average time that is spent in the susceptible class is 5 years) we have

$$\lambda = \frac{1}{5} = 0.2.$$

For a recovery rate of 10 days,

$$\nu = \left(\frac{10}{365}\right)^{-1} = 36.5$$

In R, the object `parameters` is used to store the parameter values.

```
## lambda      v
##      0.2    36.5
```

We assume a population size of $N = 5000$ with one infected individual at age=0, $I(0) = 1$. This implies that $S(0) = N - 1 = 4999$ and $R(0) = 0$. The `state` vector is given by

```
##      X      Y      Z
## 4999      1      0
```

Note that the R objects `X`, `Y` and `Z` correspond to the compartments S, I and R, respectively.

The HTML file

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er_prog1_STEAM3D_2026-V1.html Open in Browser Find Publish

1. The SIR model

2. The SIR model in a closed population

Example 1 ($\lambda = 0.2, D=10$ days)

Example 2 ($\lambda = 0.2, D=60$ days)

Example 3: age dependent force of infection

3. The SIR model in open population

4. Threshold and Equilibrium Values

$$\frac{dR(a)}{da} = \nu I(a) - \mu(a)R(a),$$

2. The SIR model in a closed population

We consider a closed population, i.e., individuals do not enter or exit to/from the population, that is $\mu(a) = 0$. In this case the SIR model is given by

$$\begin{aligned}\frac{dS(a)}{da} &= -\lambda(a)S(a), \\ \frac{dI(a)}{da} &= \lambda(a)S(a) - \nu I(a), \\ \frac{dR(a)}{da} &= \nu I(a).\end{aligned}$$

The model

There are two parameters in the model, the force of infection λ and the recovery rate ν .

Model assumptions

- No mortality access due to the disease.
- Life long immunity.

Example 1 ($\lambda = 0.2, D=10$ days)

Show

SIR model with $\lambda = 0.2$ and $D = 10$ days ($\nu = 36.5$)

For a population in which the average age at infection is 5 years (i.e., the average time that is spent in the susceptible class is 5 years) we have

$$\lambda = \frac{1}{D} = 0.2$$

21°C
Deels zonnig

Search

ENG

15:26

26/09/2026

24

The HTML file: example 1

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er_prog1_STEAM3D_2026-V1.html Open in Browser Find Publish

- Life long immunity.

1. The SIR model

2. The SIR model in a closed population

Example 1 ($\lambda = 0.2, D=10$ days)

Example 2 ($\lambda = 0.2, D=60$ days)

Example 3: age dependent force of infection

3. The SIR model in open population

4. Threshold and Equilibrium Values

Example 1 ($\lambda = 0.2, D=10$ days)

Show

SIR model with $\lambda = 0.2$ and $D = 10$ days ($\nu = 36.5$)

For a population in which the average age at infection is 5 years (i.e., the average time that is spent in the susceptible class is 5 years) we have

$$\lambda = \frac{1}{5} = 0.2.$$

For a recovery rate of 10 days,

$$\nu = \left(\frac{10}{365}\right)^{-1} = 36.5$$

In R, the object `parameters` is used to store the parameter values.

```
parameters <- c(lambda = 0.2, v=36.5)
parameters
```

Code

```
## lambda      v
##      0.2    36.5
```

Hide

We assume a population size of $N = 5000$ with one infected individual at age=0, $I(0) = 1$. This implies that $S(0) = N - 1 = 4999$ and $R(0) = 0$. The `state` vector is given by

Show

21°C Deels zonnig

Search

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The HTML file: the model in R

1. The SIR model

2. The SIR model in a closed population

Example 1 ($\lambda = 0.2, D=10$ days)

Example 2 ($\lambda = 0.2, D=60$ days)

Example 3: age dependent force of infection

3. The SIR model in open population

4. Threshold and Equilibrium Values

Model formulation

The SIR model

$$\begin{aligned}\frac{dS(a)}{da} &= -\lambda(a)S(a), \\ \frac{dI(a)}{da} &= \lambda(a)S(a) - \nu I(a), \\ \frac{dR(a)}{da} &= \nu I(a),\end{aligned}$$

can be defined in R using the `SIR` function in the following way

```
SIR<-function(t,state,parameters)
{
  with(as.list(c(state, parameters)),
  {
    dX <- -lambda*X
    dY <- lambda*X - v*Y
    dZ <- v*Y
    list(c(dX, dY, dZ))
  })
}
```

The object `times` is the ages that we use to integrate the equation system in order to produce the output. In our example, the age ranges from 0 to 40 years old.

```
## [1] 0.00 0.01 0.02 0.03 0.04 0.05 0.06 0.07 0.08 0.09 0.10 0.11 0.12 0.13 0.14
## [16] 0.15 0.16 0.17 0.18 0.19
```

The HTML file: running the model in R

1. The SIR model

2. The SIR model in a closed population

Example 1 ($\lambda = 0.2, D=10$ days)

Example 2 ($\lambda = 0.2, D=60$ days)

Example 3: age dependent force of infection

3. The SIR model in open population

4. Threshold and Equilibrium Values

Using the `ode()` function in the `deSolve` package for numerical integration.

Running the model

We solve the equations system and produce the model's solution using the function `ode` of the R package `deSolve`. In the panel below,

$$S(0) = 4999, I(0) = 1, R(0) = 0,$$

and

$$S(0.05) = 4949.259, I(0.05) = 22.989501, R(0.05) = 27.751380.$$

Numerical solution

The R object `out` contain the numerical solution of the model. The objects `X`, `Y` and `Z` are the number of susceptible, infected and immune individuals at age `a` (the R object `times`).

```
require(deSolve)
out <- as.data.frame(ode(y=state, times=times, func=SIR, parms=parameters))
head(out)
```

##	time	X	Y	Z
## 1	0.00	4999.000	1.000000	0.000000
## 2	0.01	4989.012	9.061818	1.926190
## 3	0.02	4979.044	14.641580	6.314481
## 4	0.03	4969.096	18.498345	12.405853
## 5	0.04	4959.168	21.159066	19.673391
## 6	0.05	4949.259	22.989501	27.751380

Graphical solution

Figure 1 shows the numerical solution for the example. We notice that, the fraction infected individuals is relatively small compared to the fraction susceptible and immune. This is due to the different time spent in each compartment. The average duration in the susceptible class is 5 year and after recovery individuals gain life long immunity against reinfection.

The HTML file

1. The SIR model

2. The SIR model in a closed population

Example 1 ($\lambda = 0.2, D=10$ days)

Example 2 ($\lambda = 0.2, D=60$ days)

Example 3: age dependent force of infection

3. The SIR model in open population

4. Threshold and Equilibrium Values

Graphical solution

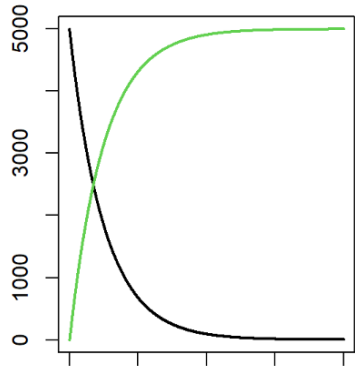
Figure 1 shows the numerical solution for the example. We notice that, the fraction infected individuals is relatively small compared to the fraction susceptible and immune. This is due to the different time spent in each compartment. The average duration in the susceptible class is 5 year and after recovery individuals gain life long immunity against reinfection.

```
par(mfrow=c(1,2), oma=c(0,0,3,0))
plot (times,out$X ,type="l",main="S and R", xlab="age", ylab="-",lwd=2)
lines (times,out$Z,col=3,lwd=2)
plot (times,out$Y ,type="l",main="I", xlab="time", ylab="-",lwd=2)
mtext(outer=TRUE,side=3,"SIR model, D=10 days",cex=1.5)
```

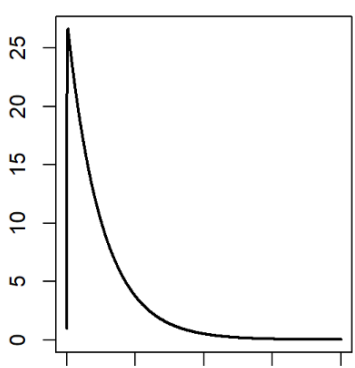
Code to produce the figures.

SIR model, D=10 days

S and R



I

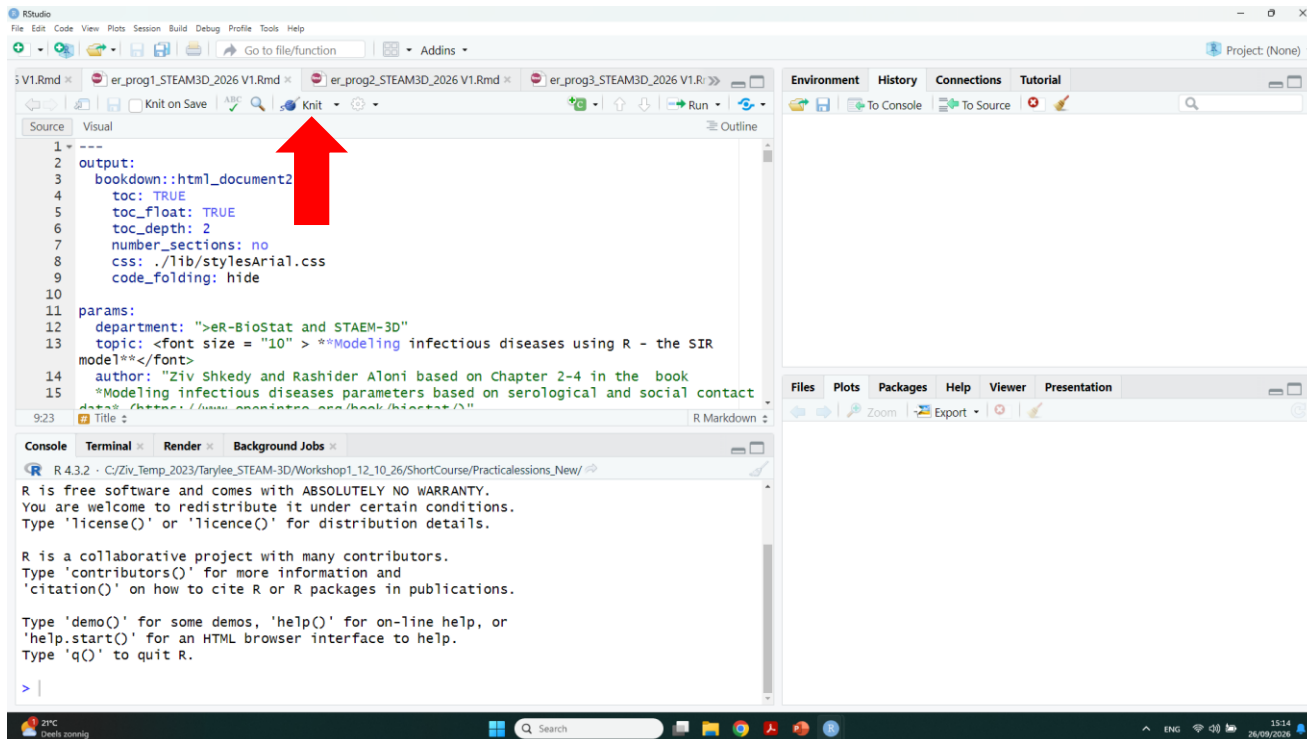


Running the model in R

- Produce the model's output by yourself.
- Open R/R studio.
- Open the program:
 - [ModelingIDinR1_V1_Stat&Dynam_10_2026.R.](#)

How to produce the HTML file ?

- Open R Studio.
- Open the file: `eR_prog1_STEAM3D_2026_V1.Rmd`.
- Knit the file.



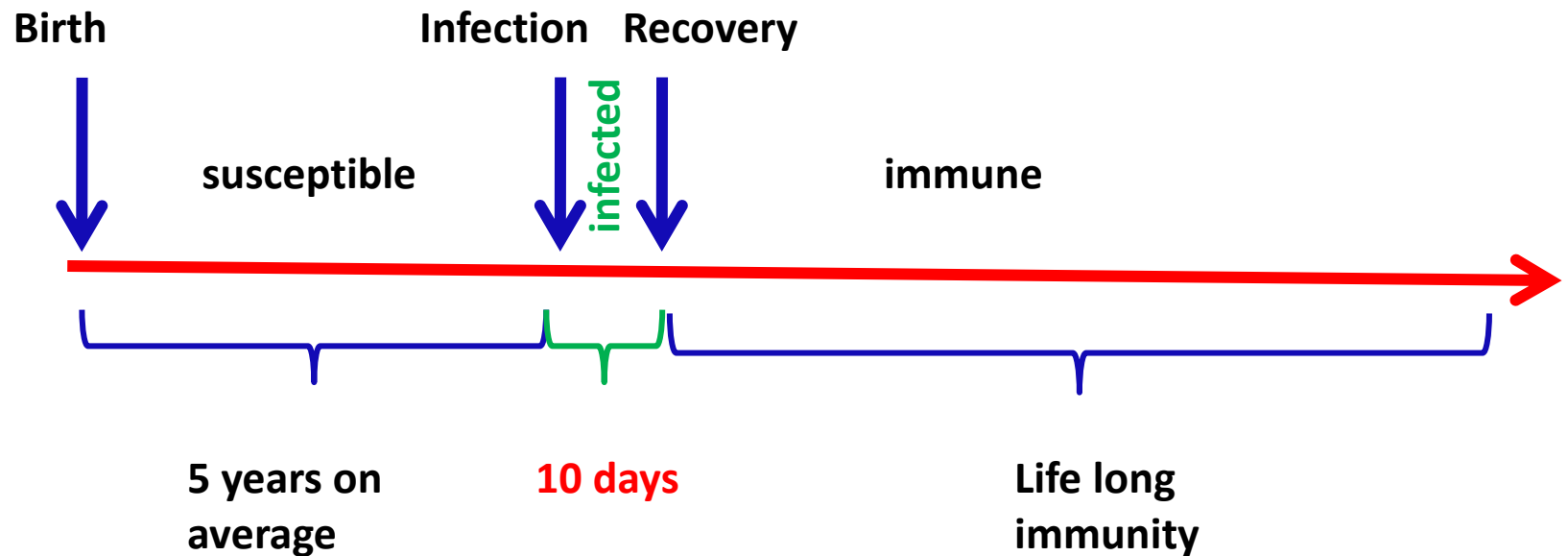
Changing the parameter setting

- We run the model again but higher duration in the susceptible class.

$$D=10 \text{ days} \rightarrow D = 60 \text{ days}$$

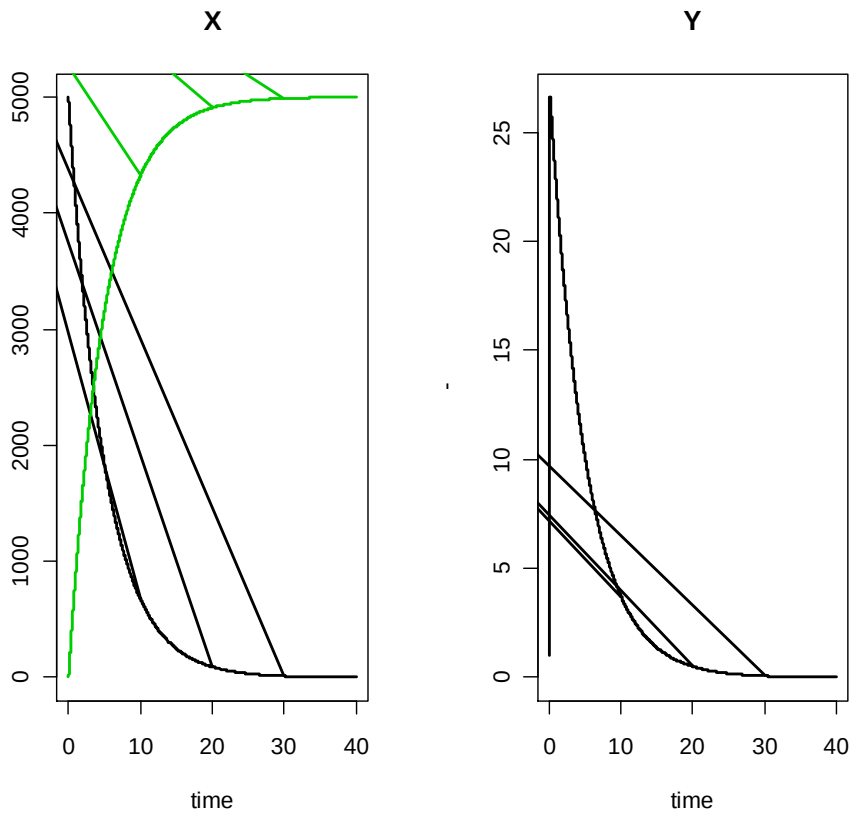
$$\left(\frac{10}{365}\right)^{-1} = 36.5 \quad \left(\frac{60}{365}\right)^{-1} = 6.08$$

Duration of stay in the different compartments of the models



Graphical output

SIR model



We expect to see only few infected individuals at each age (compared to the number of susceptible and immune).

Slides 31-37: example 2 in the HTML file changing the recovery rate

1. The SIR model

2. The SIR model in a closed population

Example 1 ($\lambda = 0.2, D=10$ days)

Example 2 ($\lambda = 0.2, D=60$ days)

Example 3: age dependent force of infection

3. The SIR model in open population

4. Threshold and Equilibrium Values

Figure 1: Solution for a SIR model with $\lambda=0.2$ and $D=10$ days.

Example 2 ($\lambda = 0.2, D=60$ days)

SIR with $\lambda = 0.2$ $D=\text{two months}$ ($\nu = 6.08$)

In this example we change the recovery rate from 10 days to two months (60 days) so individuals stay in the infected class a longer time compare to the first example. In this case

$$\nu = \left(\frac{60}{365}\right)^{-1} = 6.08$$

Show

[1] 0.1643836

Show

[1] 6.083333

The new parameters vector is given by

Show

lambda v
0.200000 6.083333

Show

Runing the model

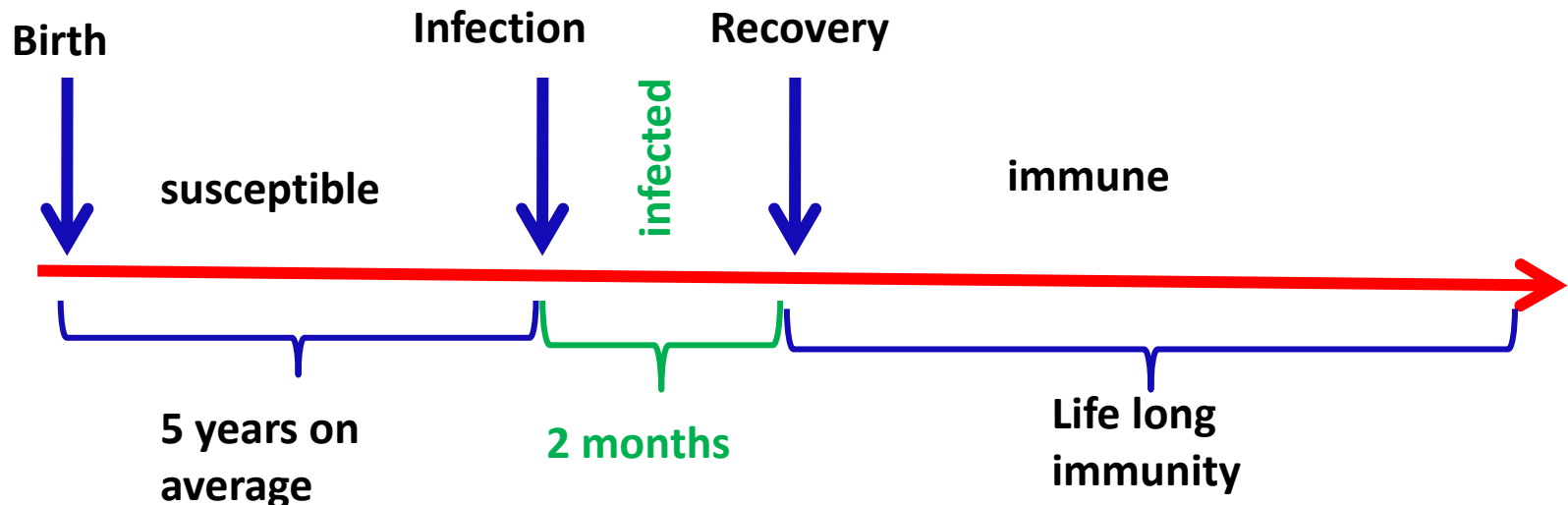
Note that compared to an average duration of 10 days in the infected class (Example 1), we expect to observed a high proportion of

$D=10 \text{ days} \rightarrow D = 60 \text{ day}$

Duration of stay in the different compartments of the models

Let us assume that the recovery rate is 2 months (i.e. individuals stay in the infected class 2 months)

What do we expect to see ?



The model with recovery rate of 2 months

1/(60/365)

```
parameters <- c(lambda = 0.2, v=6.083333)
```

```
parameters
```

```
state <- c(X=4999,Y=1,Z=0)
```

```
state
```

```
SIR<-function(t,state,parameters)
```

```
{
```

```
with(as.list(c(state, parameters)),
```

```
{
```

```
dX <- -lambda*X
```

```
dY <- lambda*X - v*Y
```

```
dZ <- v*Y
```

```
list(c(dX, dY, dZ))
```

```
})
```

```
}
```

```
times<-seq(0,40,by=0.01)
```

```
times
```

```
require(deSolve)
```

```
out <-
```

```
as.data.frame(ode(y=state,times=times,func=SIR,parms=parameters))
```

```
head(out)
```

```
par(mfrow=c(1,2), oma=c(0,0,3,0))
```

```
plot(times,out$X,type="l",main="X", xlab="time",  
ylab="-",lwd=2)
```

```
lines(times,out$Z,col=3,lwd=2)
```

```
plot(times,out$Y,type="l",main="Y", xlab="time",  
ylab="-",lwd=2)
```

```
mtext(outer=TRUE,side=3,"SIR model",cex=1.5)
```

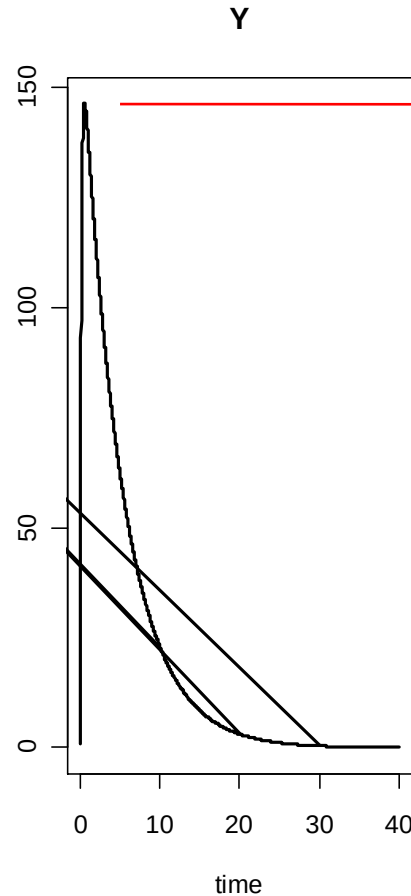
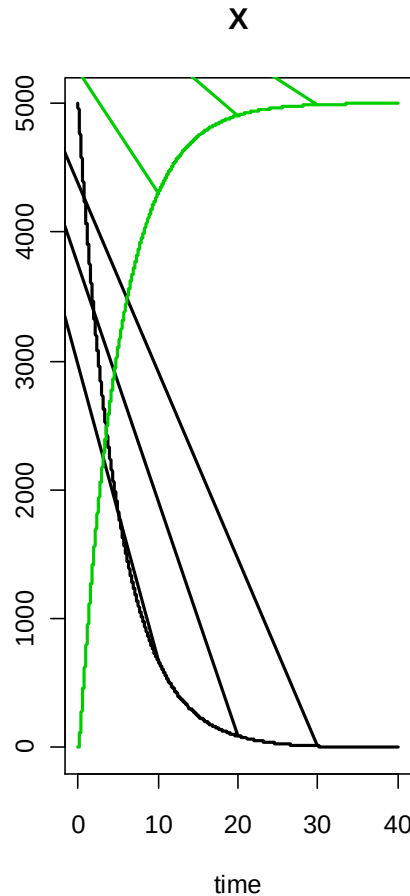

$$\left(\frac{60}{365}\right)^{-1} = 6.08$$

- Recover rate of 2 months
- (60 days).
- Code from the R program:
 - [ModelingIDinR1_V1_Stat&Dynam_10_2026.R](#)

Graphical output

(force of infection of 0.2 and recovery rate of 2 months)

SIR model



~ 150 infected individuals at the peak (compare with ~25 for recovery rate of 10 days)

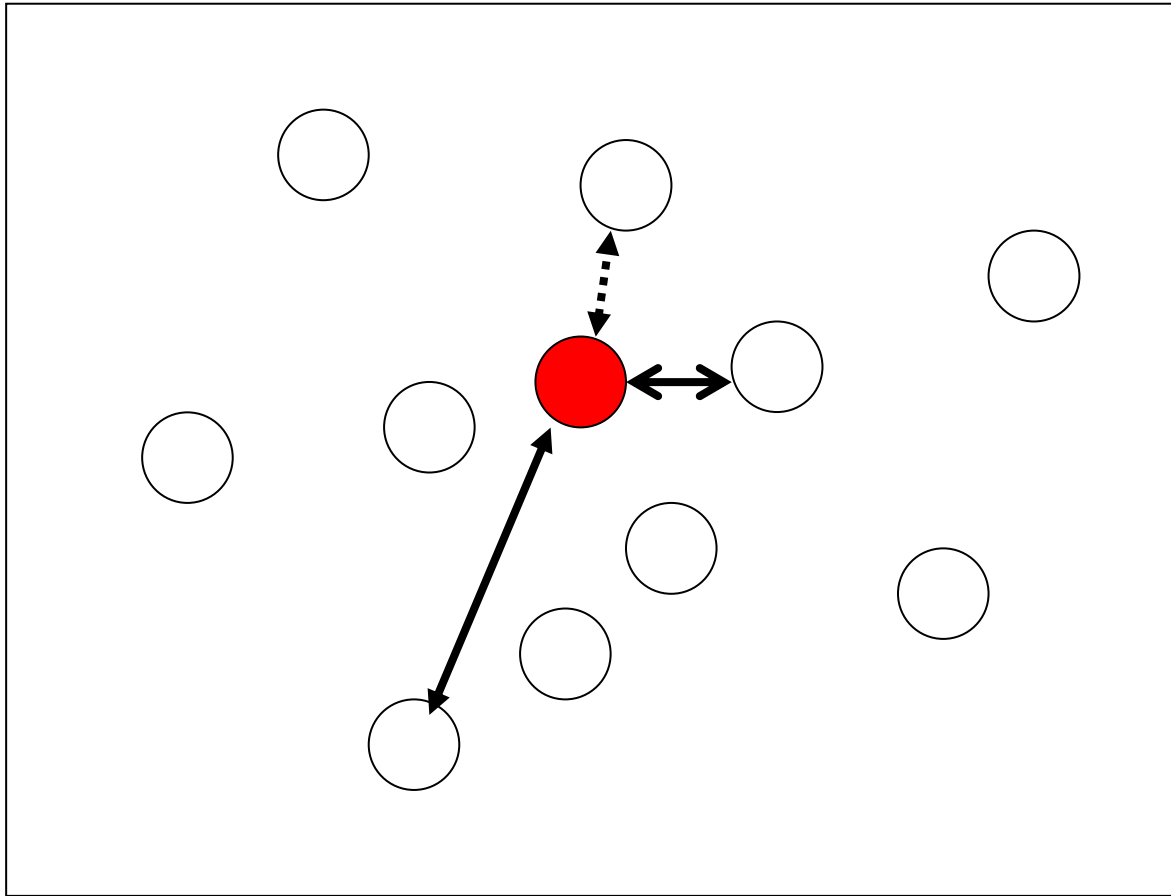
Running the model in R

- Produce the model's output by yourself.
- Open R/R studio.
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Part 2:

The mass action principle (static model)

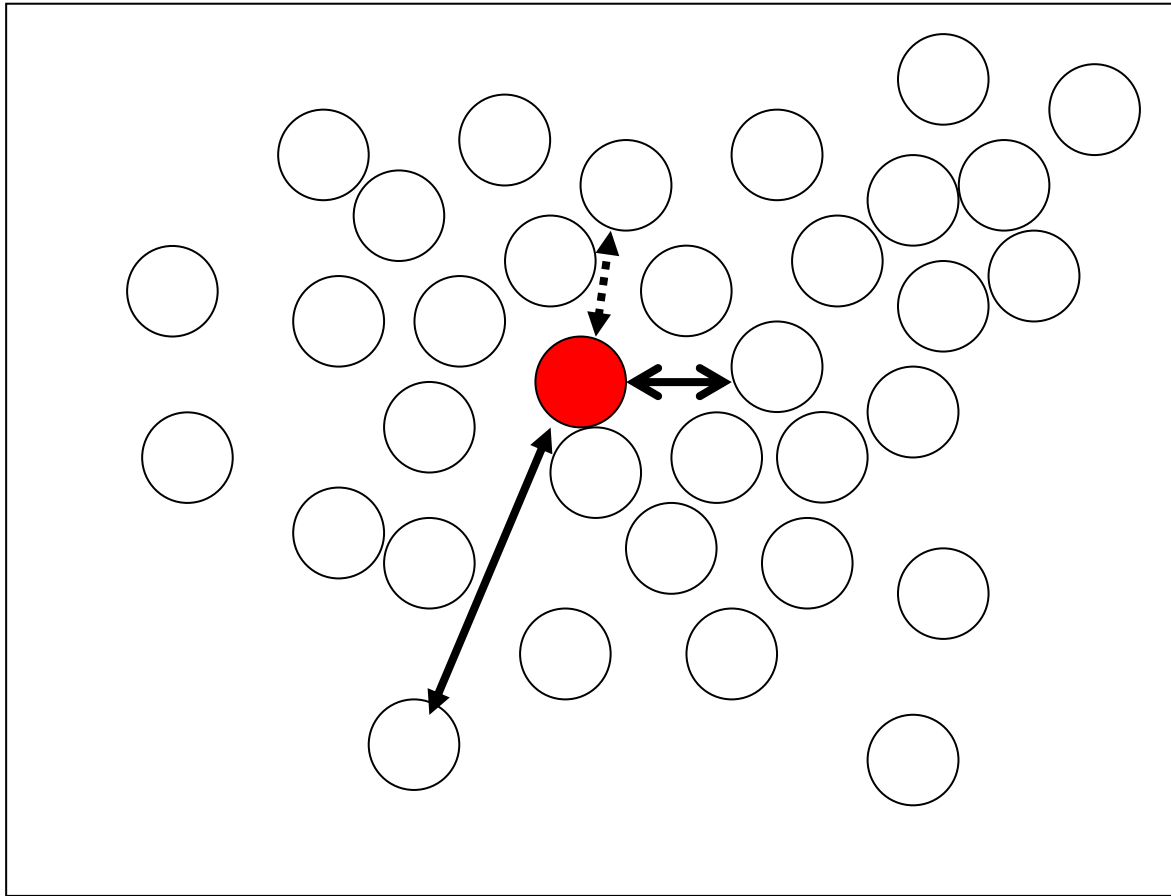
The Mass-Action Principle



Contacts are made in random.

Number of new cases = $P(\text{transmission}) \times \# \text{ of infectious} \times \# \text{ of susceptible}$

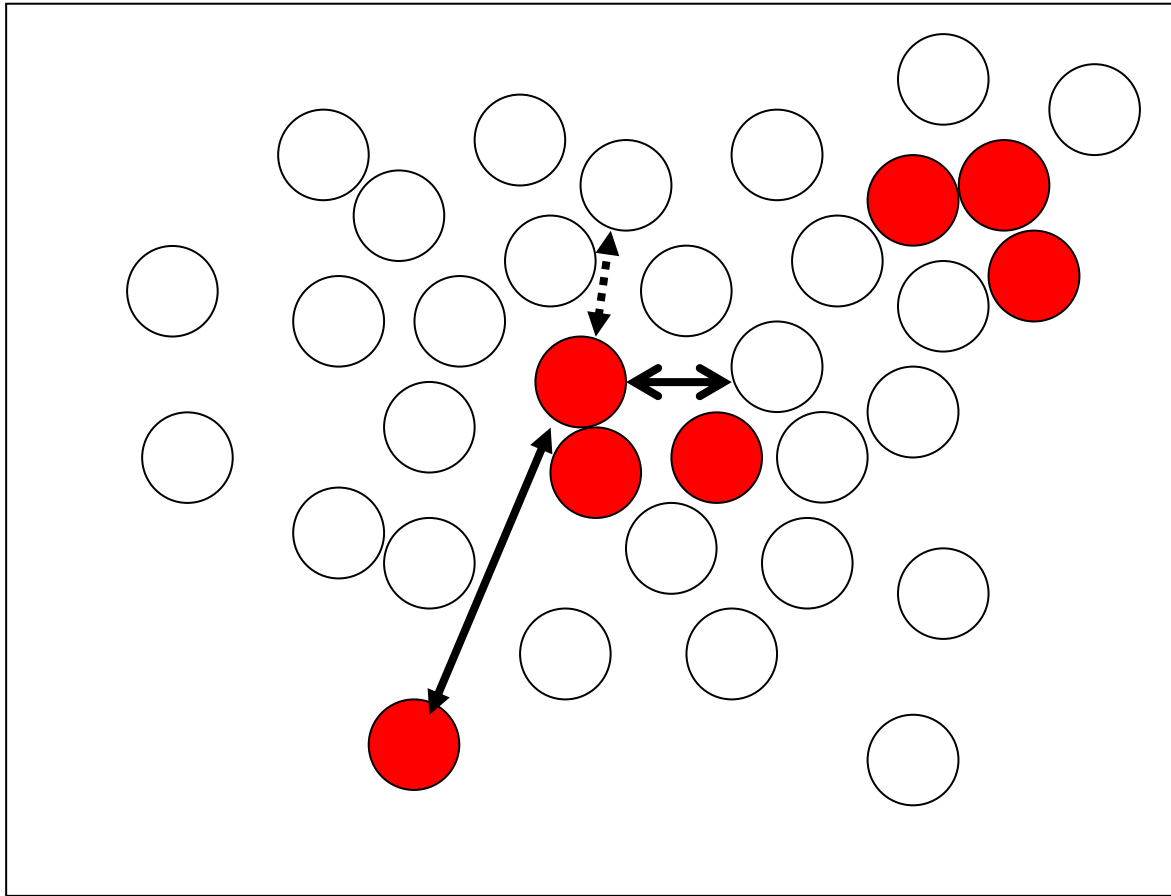
The Mass-Action Principle



Contacts are made in random.

Number of new cases = $P(\text{transmission}) \times \# \text{ of infectious} \times \# \text{ of susceptible}$

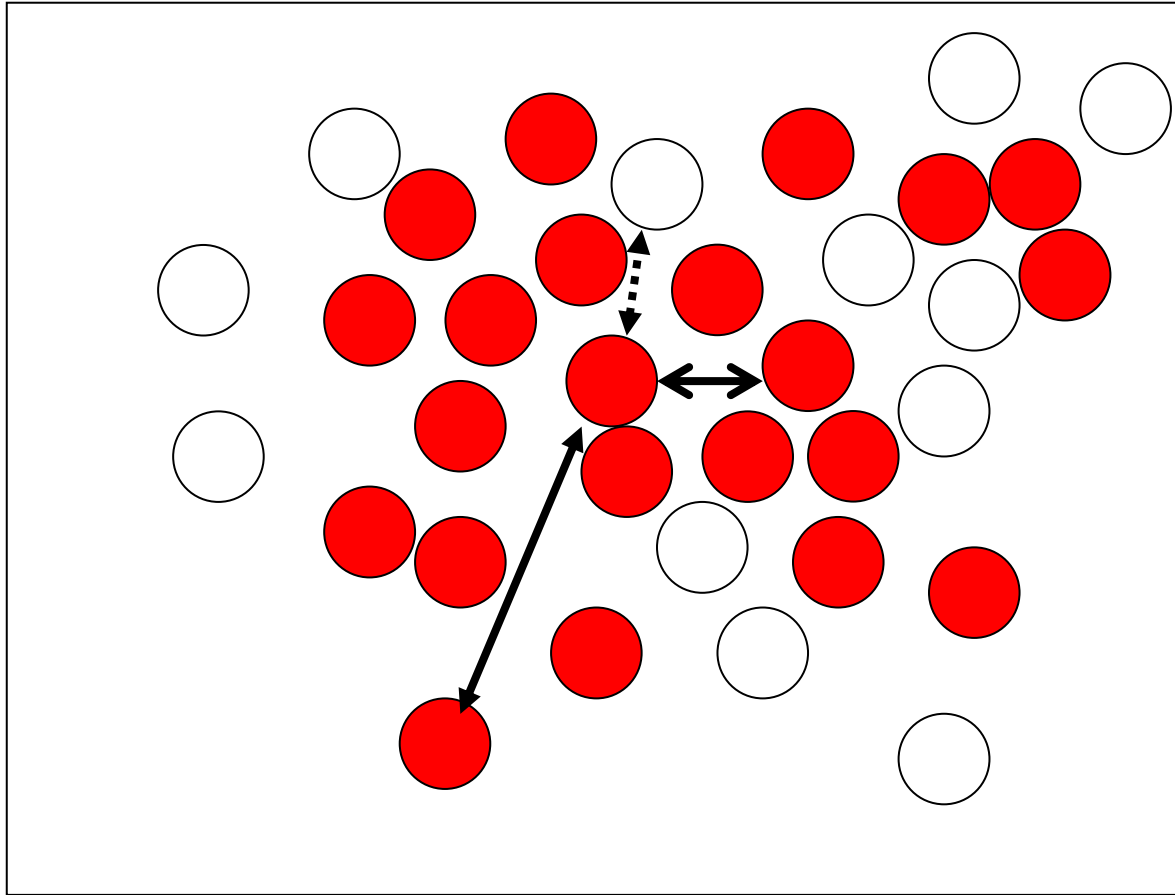
The Mass-Action Principle



Contacts are made in random.

Number of new cases = $P(\text{transmission}) \times \# \text{ of infectious} \times \# \text{ of susceptible}$

The Mass-Action Principle



Contacts are made in random.

Number of new cases:

$$\beta \times I \times S$$

Transmission
probability per
contact

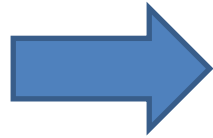
Number of new cases = $P(\text{transmission}) \times \# \text{ of infectious} \times \# \text{ of susceptible}$

The Mass-Action Principle

$$\frac{dS(a)}{da} = -\lambda S(a)$$

$$\frac{dI(a)}{da} = \lambda S(a) - \sigma I(a)$$

$$\frac{dR(a)}{da} = \sigma I(a)$$



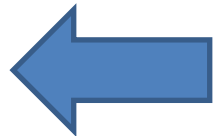
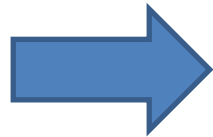
$$\frac{dS(a)}{da} = -\beta \times I(a) \times S(a)$$

$$\frac{dI(a)}{da} = \beta \times I(a) \times S(a) - \sigma I(a)$$

$$\frac{dR(a)}{da} = \sigma I(a)$$

Age dependent force of infection

$$\begin{aligned}\frac{dS(a)}{da} &= -\lambda(a)S(a) \\ \frac{dI(a)}{da} &= \lambda(a)S(a) - \sigma I(a) \\ \frac{dR(a)}{da} &= \sigma I(a)\end{aligned}$$



$$\begin{aligned}\frac{dS(a)}{da} &= -\beta \times I(a) \times S(a) \\ \frac{dI(a)}{da} &= \beta \times I(a) \times S(a) - \sigma I(a) \\ \frac{dR(a)}{da} &= \sigma I(a)\end{aligned}$$

$$\lambda(a) = \beta \times I(a)$$

Age
dependent

constant

Model parameters

$$\frac{dS(a)}{da} = -\beta \times I(a) \times S(a)$$
$$\frac{dI(a)}{da} = \beta \times I(a) \times S(a) - \sigma I(a)$$
$$\frac{dR(a)}{da} = \sigma I(a)$$

Parameters in R:

```
> parameters <- c(beta=0.0085, v=36.5)
> state <- c(X=4999, Y=1, Z=0)
> parameters
  beta      v
0.0085 36.5000
> state
  X    Y    Z
4999  1    0
```

$$\beta = 0.0085$$
$$\sigma = 36.5 \quad (10 \text{ days})$$

10 days

$$\frac{10}{365} = 10 \text{ days in years}$$

$$\left(\frac{10}{365}\right)^{-1} = \left(\frac{365}{10}\right) = 36.5 = \sigma$$

Recovery rate= (duration in the infected class in years)⁻¹

Specification of the model in R

$$\begin{aligned}\frac{dS(a)}{da} &= -\beta \times I(a) \times S(a) \\ \frac{dI(a)}{da} &= \beta \times I(a) \times S(a) - \sigma I(a) \\ \frac{dR(a)}{da} &= \sigma I(a)\end{aligned}$$

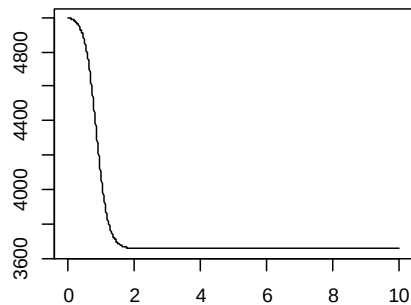


```
SIR<-function(t,state,parameters)
{
  with(as.list(c(state,
parameters))),
  {
    dX <- -beta*Y*X
    dY <- beta*Y*X - v*Y
    dZ <- v*Y
    list(c(dX, dY, dZ))
  })
}
```

Numerical solution

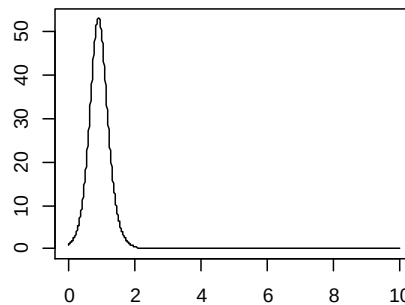
SIR model, $\lambda = \beta \cdot I$

X



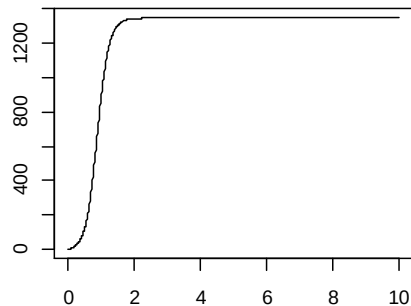
time

Y



time

Z



time

After ~ 2 years there are no infected individuals.

~ 3600 individuals will not be infected.

Equilibrium

Slides 39-48: example 3 in the HTML file

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Figure 2: Solutions for a SIR model D=60 days.

1. The SIR model
2. The SIR model in a closed population
Example 1 ($\lambda = 0.2, D=10$ days)
Example 2 ($\lambda = 0.2, D=60$ days)
Example 3: age dependent force of infection
3. The SIR model in open population
4. Threshold and Equilibrium Values

Example 3: age dependent force of infection

In this example we change the formulation of the force of infection and assume that $\lambda(a) = \beta \text{ times } I(a)$. Here β is the transmission rate and $I(a)$ is the number of infected individuals at age a .

$$\beta = 0.0085 \text{ and } \lambda = \beta \times I(a)$$

For a recovery rate of 10 days, the new parameters vector is given by

Show

```
##      beta      v
## 0.0085 36.5000
```

Show

Definition of the model

For the SIR model, we use $\beta \times Y \times X$ instead of $\lambda \times X$

Show

Runing the model and graphical output

The solution of the model is presnted in Figure XXX. Note that the force of infection in this model is not constant but it is proportional for the number of individuals in the infected compartment presented in Figure 3.

Show

```
##      time      X      Y      Z
## 1 0.00 4999.000 1.000000 0.000000
## 2 0.01 4998.562 1.061727 0.3761535
## 3 0.02 4998.097 1.127220 0.7755187
```

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The HTML file: the model

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er_prog1_STEAM3D_2026-V1.html Open in Browser

Find

Publish

1. The SIR model

2. The SIR model in a closed population

Example 1 ($\lambda = 0.2, D=10$ days)

Example 2 ($\lambda = 0.2, D=60$ days)

Example 3: age dependent force of infection

3. The SIR model in open population

4. Threshold and Equilibrium Values

Definition of the model

For the SIR model, we use $\beta Y X$ instead of $\lambda Y X$

```
SIR<-function(t,state,parameters)
{
  with(as.list(c(state, parameters)),
  {
    dX <- -beta*Y*X
    dY <- beta*Y*X - v*Y
    dZ <- v*Y
    list(c(dX, dY, dZ))
  })
}
```

Code for the model.

Runing the model and graphical output

The solution of the model is presnted in Figure XXX. Note that the force of infection in this model is not constant but it is proportional for the number of individuals in the infected compartment presented in Figure 3.

```
##   time      X      Y      Z
## 1 0.00 4999.000 1.000000 0.000000
## 2 0.01 4998.562 1.061727 0.3761535
## 3 0.02 4998.097 1.127220 0.7755187
## 4 0.03 4997.604 1.196705 1.1995104
## 5 0.04 4997.080 1.270419 1.6496279
## 6 0.05 4996.524 1.348611 2.1274607
```

Running the model in R

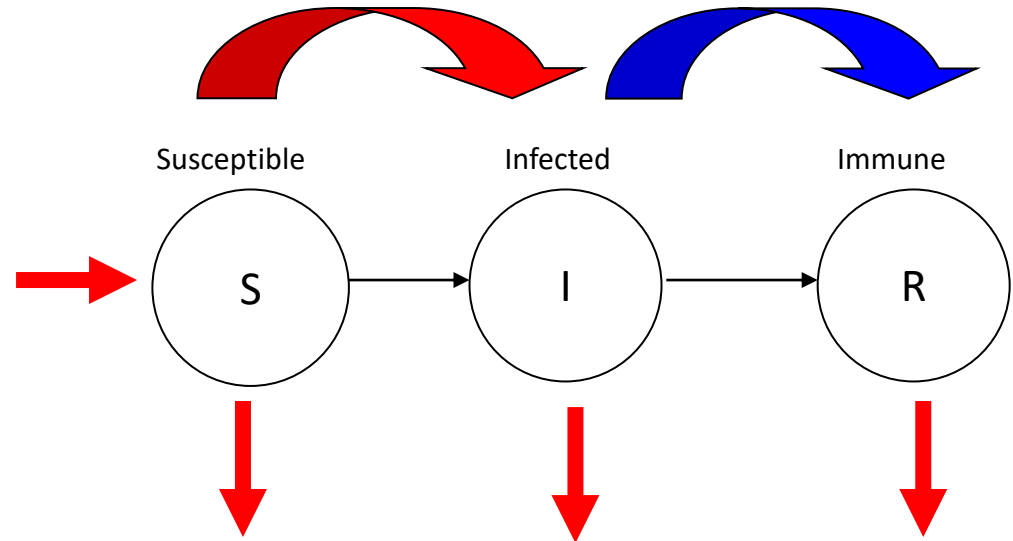
- Produce the model's output by yourself.
 - Open R/R studio.
 - Open the program:
 - [ModelingIDinR1_V1_Stat&Dynam_10_2026.R.](#)

Part 3:

The SIR model in open population

SIR transmission model in open population

$$\begin{aligned}\frac{dS(t)}{dt} &= N\mu - \lambda S - \mu S \\ \frac{dI(t)}{dt} &= \lambda S - (\sigma + \mu)I \\ \frac{dR(t)}{dt} &= \sigma I - \mu R\end{aligned}$$



Birth and death rate are equal to μ (constant population size).

Population size: N .

Recovery rate: σ .

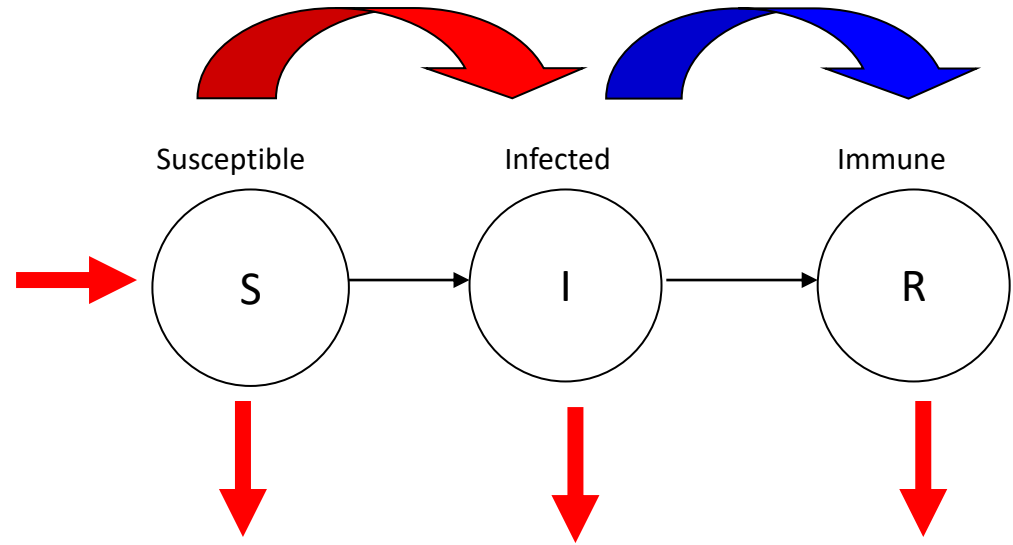
Force of infection: λ .

SIR transmission model in open population

- The model in R

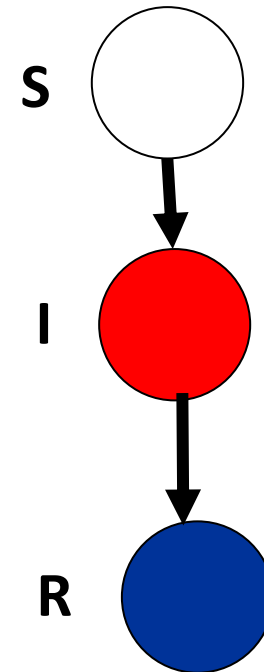
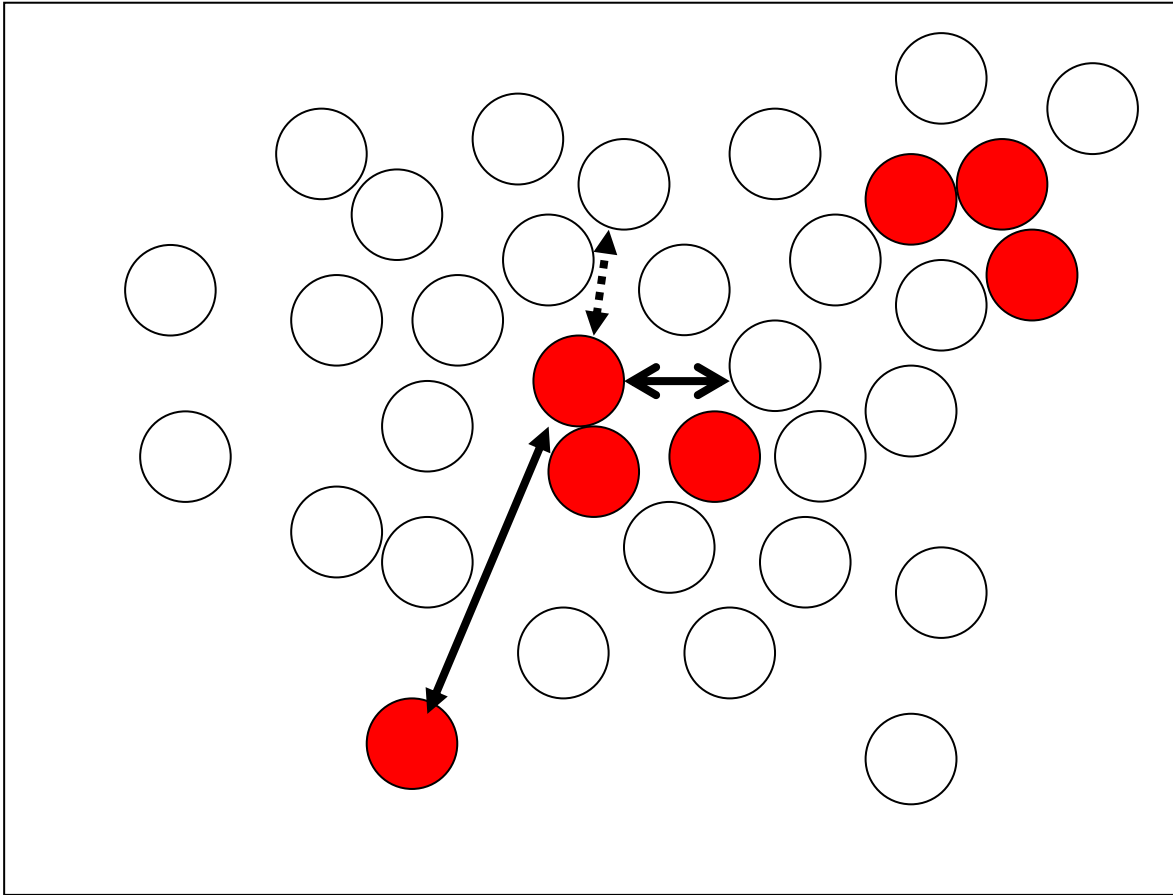
```
SIR<-function(t,state,parameters)
{
  with(as.list(c(state, parameters)),
  {
    dX <- 5000*mu-lambda*X-mu*X
    dY <- lambda*X - v*Y-mu*Y
    dZ <- v*Y-mu*Z
    list(c(dX, dY, dZ))
  })
}
```

- We assume that $N=5000$.



$$\begin{cases} \frac{dS(t)}{dt} = N\mu - \lambda S - \mu S \\ \frac{dI(t)}{dt} = \lambda S - (\sigma + \mu)I \\ \frac{dR(t)}{dt} = \sigma I - \mu R \end{cases}$$

The law of mass action



New cases(t)=transmission rate X infected(t) X susceptible(t)

SIR model in open population: Dynamic aspects

- In order to understand the dynamic of the SIR model we need to allow for time dependent force of infection.
- Open population.
- The force of infection (depends on time):

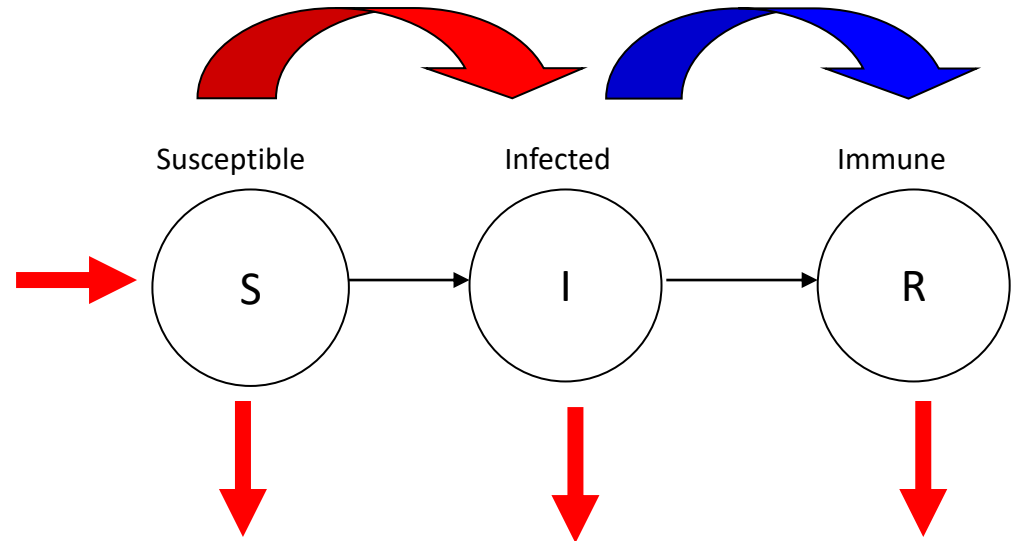
$$\lambda(t) = \beta \times I(t)$$

- Mass action principle.

$$\text{New case} = \beta \times I(t) \times S(t)$$

SIR transmission model in open population

$$\begin{aligned}\frac{dS(t)}{dt} &= N\mu - \beta IS - \mu S \\ \frac{dI(t)}{dt} &= \beta IS - (\sigma + \mu)I \\ \frac{dR(t)}{dt} &= \sigma I - \mu R\end{aligned}$$



Birth and death rate are equal to μ (constant population size).

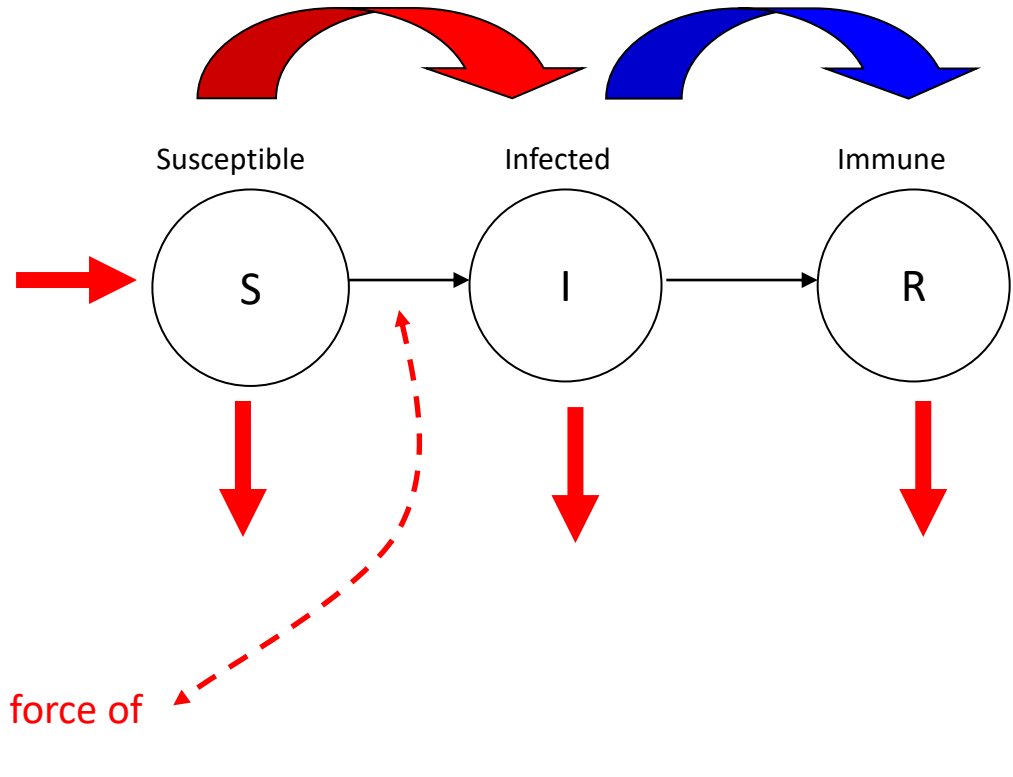
N : population size.

σ : recovery rate.

β : transmission rate.

SIR transmission model in open population

$$\begin{aligned}\frac{dS(t)}{dt} &= N\mu - \beta IS - \mu S \\ \frac{dI(t)}{dt} &= \beta IS - (\sigma + \mu)I \\ \frac{dR(t)}{dt} &= \sigma I - \mu R\end{aligned}$$

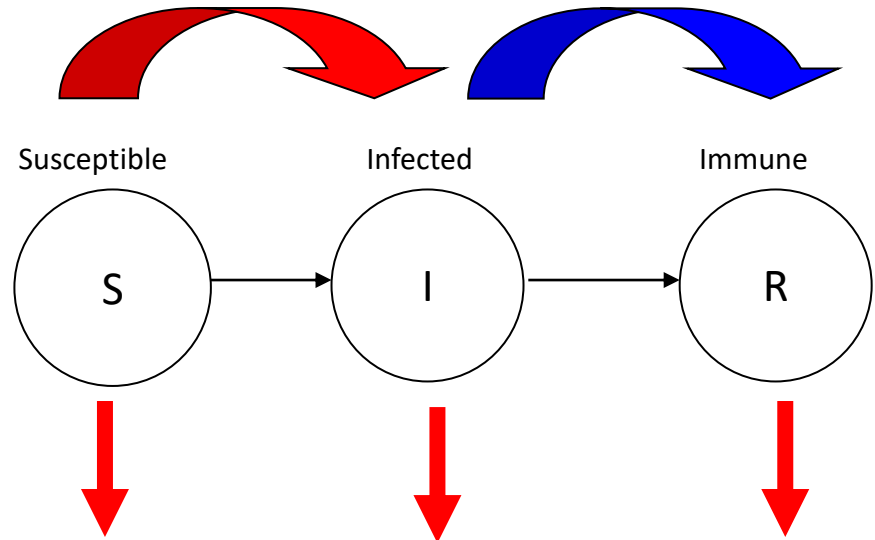


$\lambda(t) = \beta \times I(t)$: force of infection
 $\beta \times I(t) \times S(t)$: number of new cases

SIR transmission model in open population

- The model in R

```
SIR<-function(t,state,parameters)
{
  with(as.list(c(state, parameters)),
  {
    dX <- N*mu*(1-p)-beta*Y*X - mu*X
    dY <- beta*Y*X - v*Y - mu*Y
    dZ <- v*Y -mu*Z+N*mu*p
    list(c(dX, dY, dZ))
  })
}
```



- Note that $p=0$!!!

$$\left\{ \begin{array}{l} \frac{dS(t)}{dt} = N\mu - \beta IS - \mu S \\ \frac{dI(t)}{dt} = \beta IS - (\sigma + \mu)I \\ \frac{dR(t)}{dt} = \sigma I - \mu R \end{array} \right.$$

Running the model in R

```
out <-  
as.data.frame(ode(y=state, times=times, func=SIR, parms=parameters))
```

- We need to provide:
 - The state variables.
 - Time for integration.
 - The parameter setting.
 - The function for the model.

Parameters setting

- Parameter setting:

$\beta = \text{tranmission rate} = 0.001.$

$N = \text{population size} = 5000.$

$\mu = \text{birth rate} = \text{death rate} = 1/75.$

$v = \text{recovery rate} = 1.$

$p = \text{vaccinate rate} = 0.$

- Parameter setting in R:

```
> parameters <- c(mu=1/75, beta=0.001, v=1)
```

```
> print(parameters)
```

mu	beta	v
0.01333333	0.00100000	1.00000000

State variables and time for integration

- The state vector:

$$(S(0) = N - 1 = 4999, I(0) = 1, R(0) = 0)$$

- The state vector in R:

```
> state <- c(X=4999, Y=1, Z=0)
> print(state)
      X      Y      Z
4999      1      0
```

Population size and time for integration

- Population size:

```
> p<-0.0
> p
[1] 0
> N<-5000
> N
[1] 5000
```

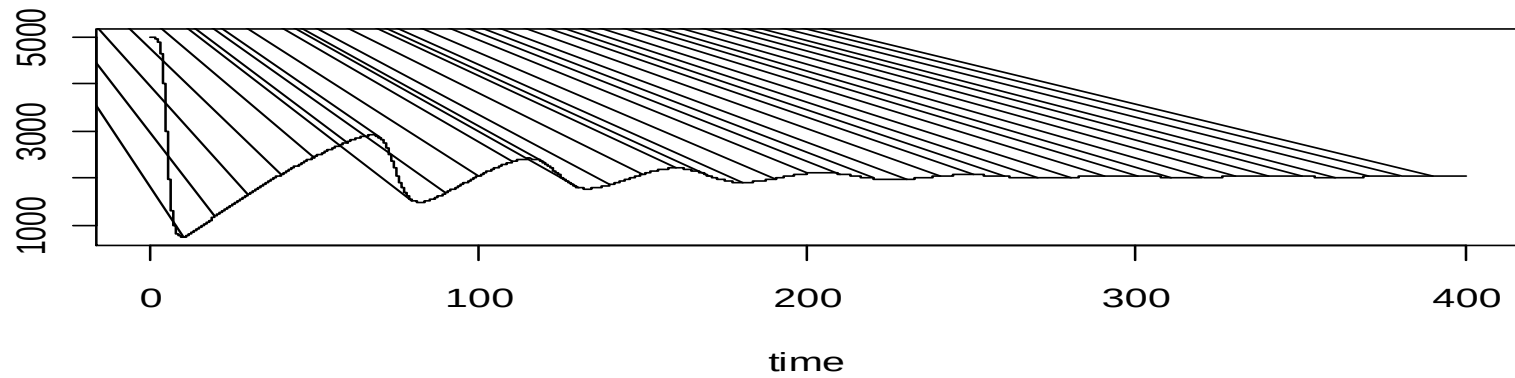
- Time for integration:

```
times<-seq(0,400,by=0.01)
print(times[1:10])
[1] 0.00 0.01 0.02 0.03 0.04 0.05 0.06 0.07 0.08
```

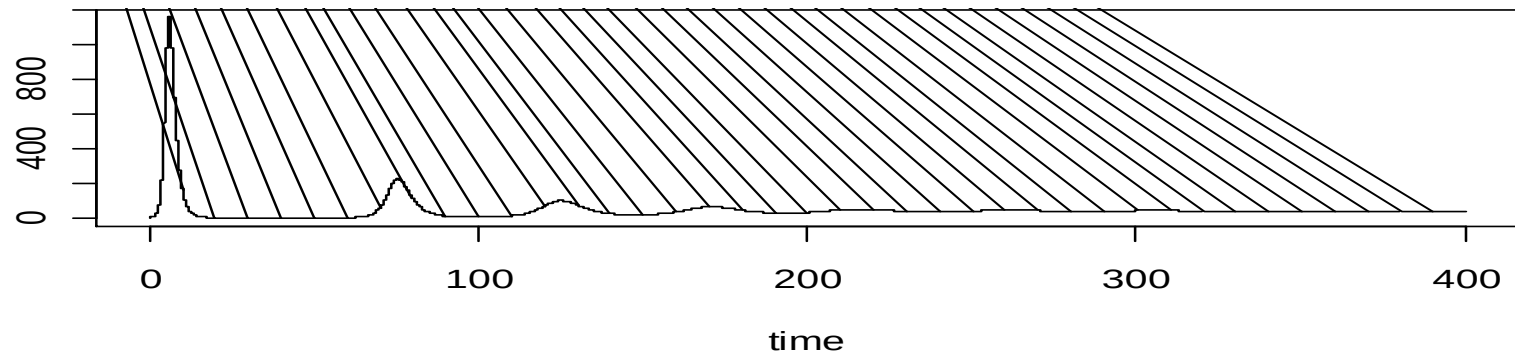
Solution for the model

SIR model

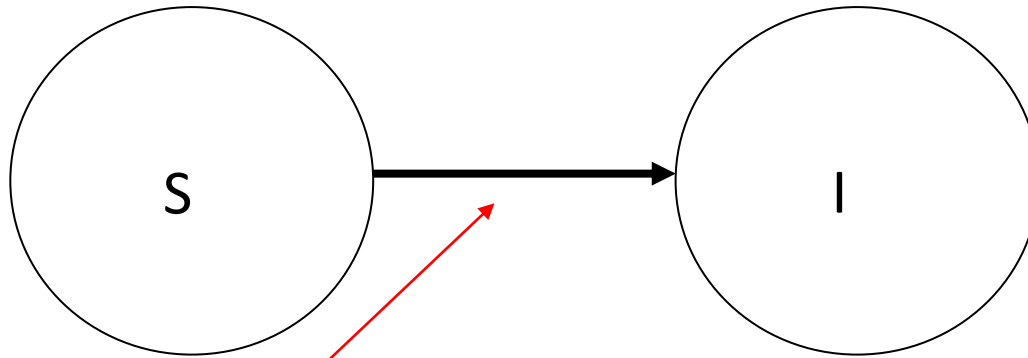
S



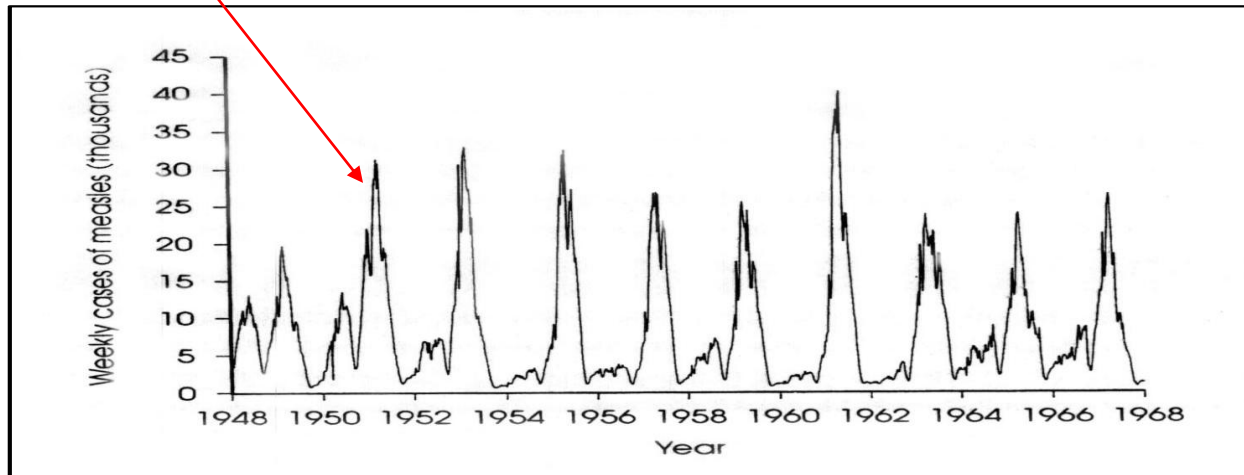
I



Incidence data : Measles in UK

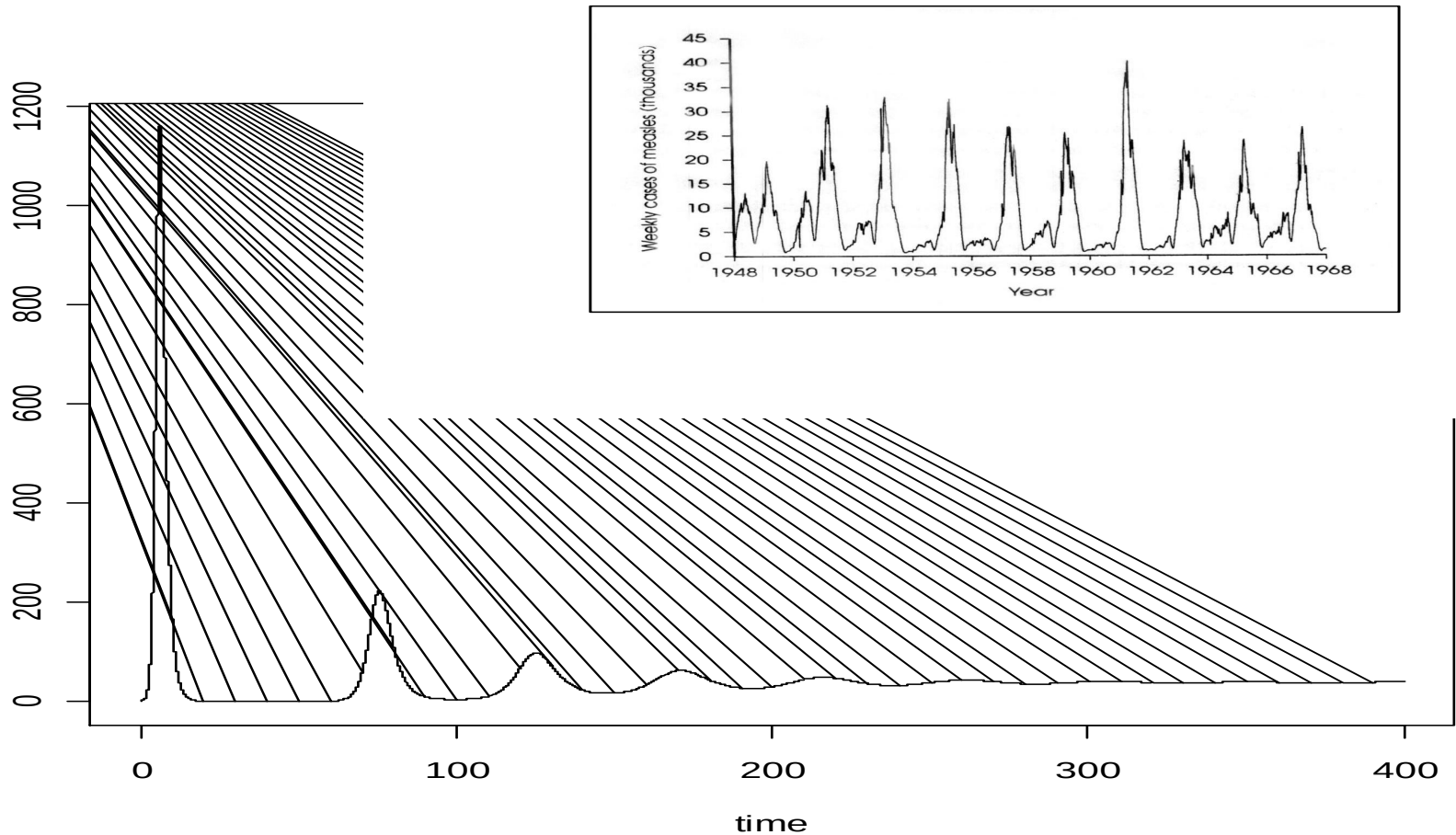


The **weekly** number of individuals who move from the susceptible to the infected class



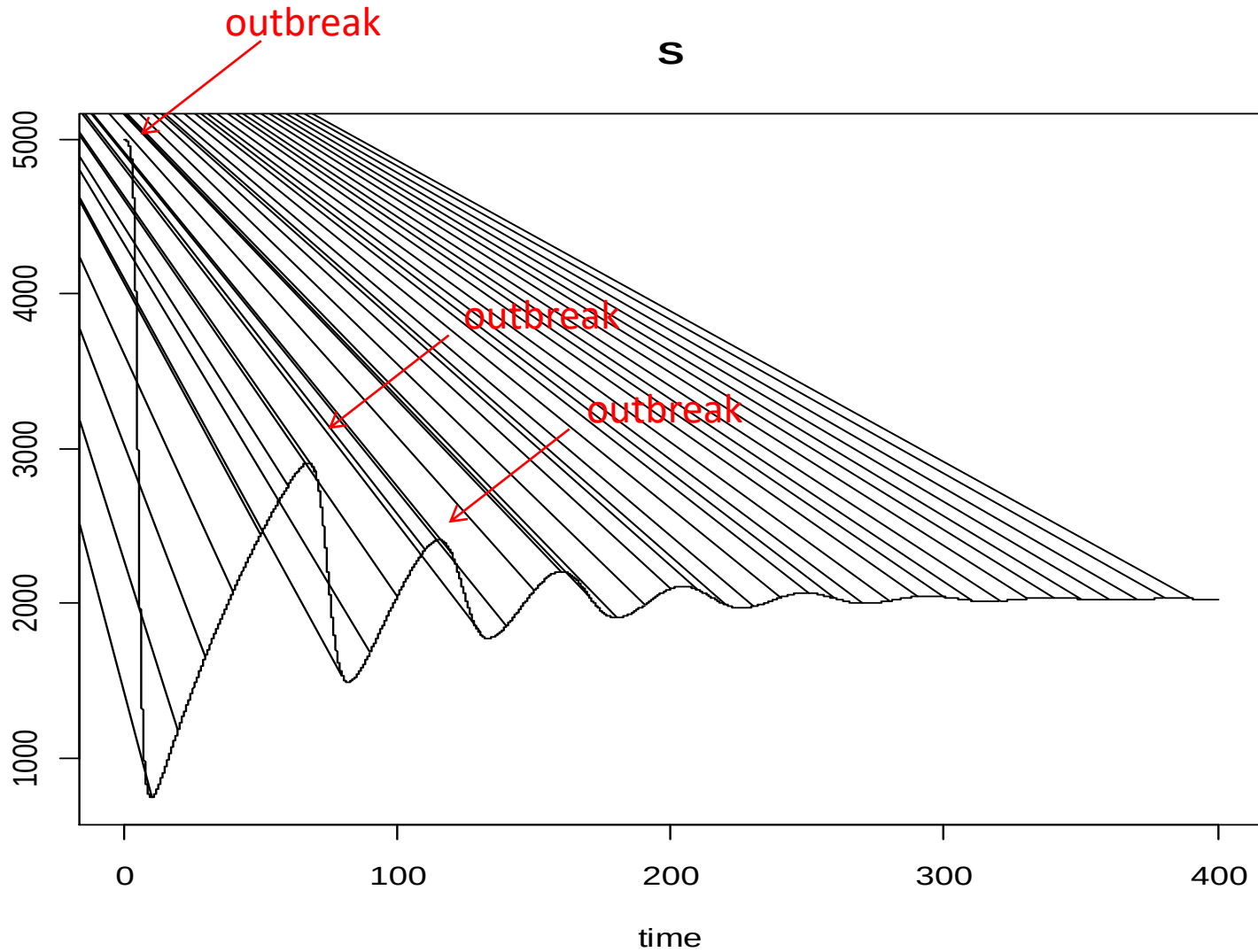
Time unit=week

Observed outbreak and predicted outbreak

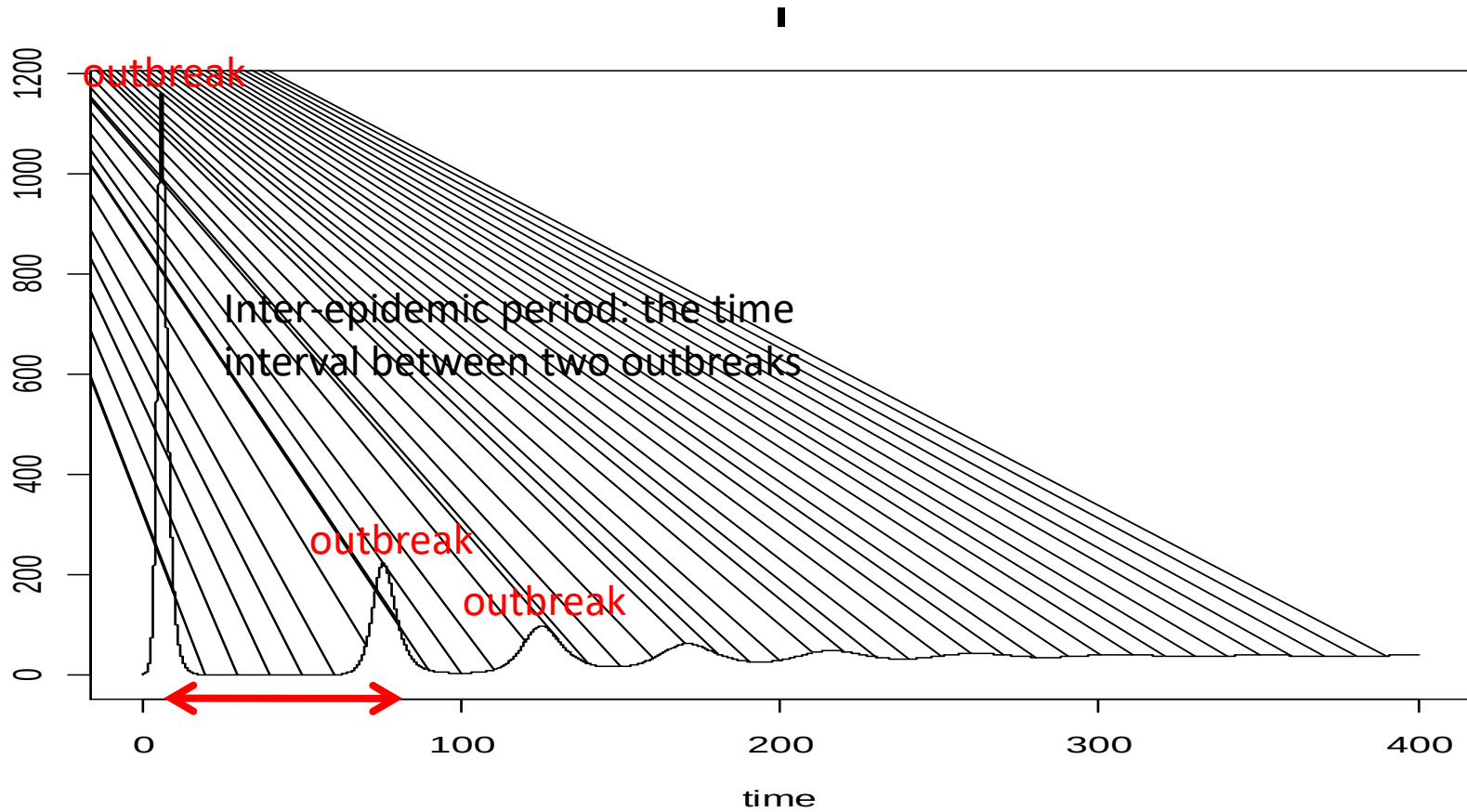


15

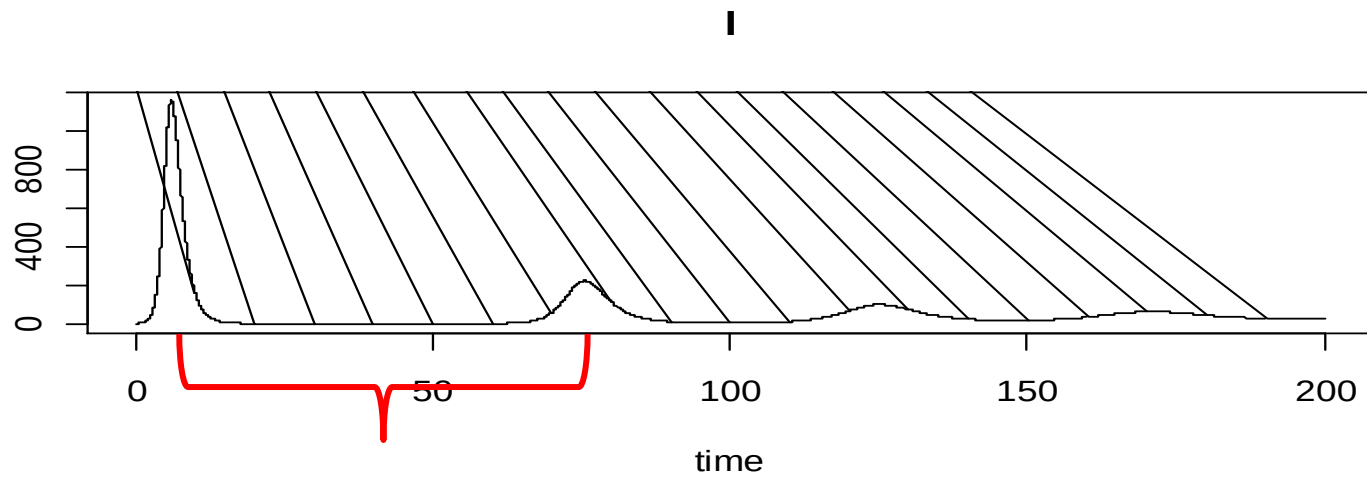
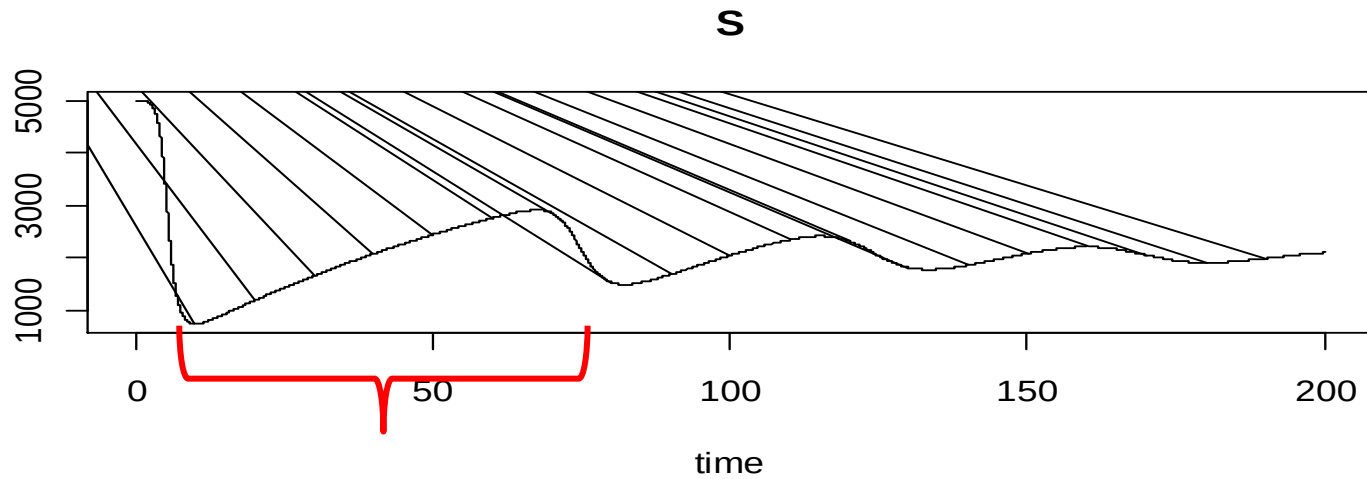
The susceptible class



The infected class

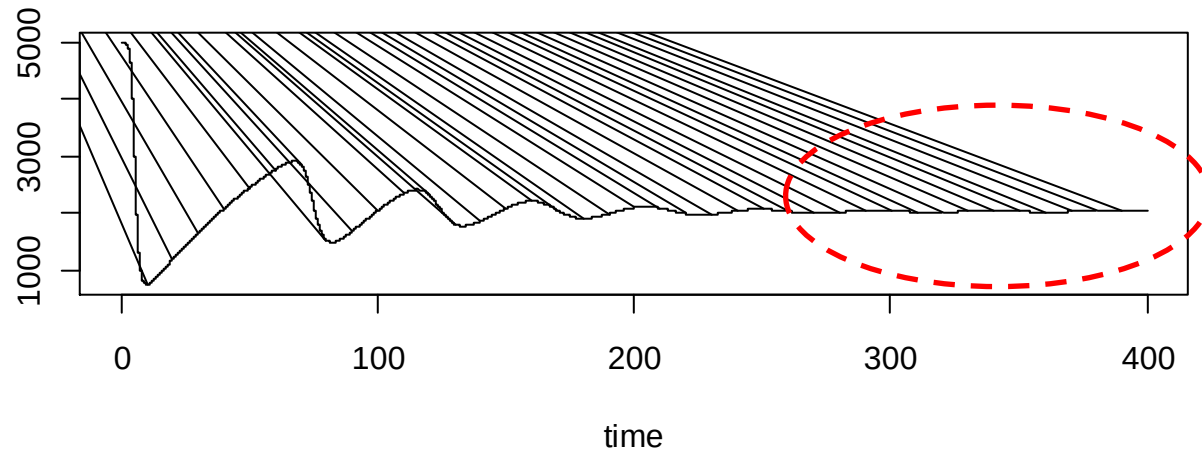


The Inter epidemic period

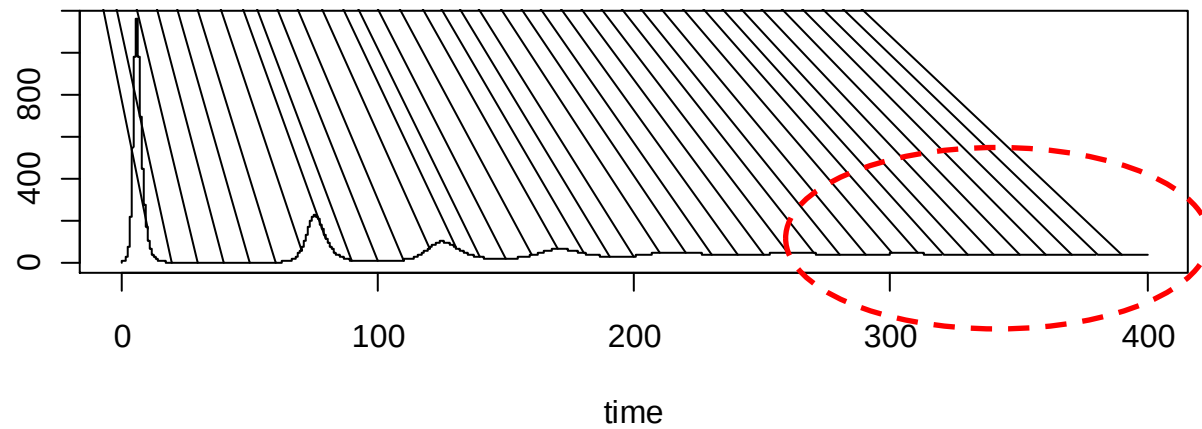


Equilibrium

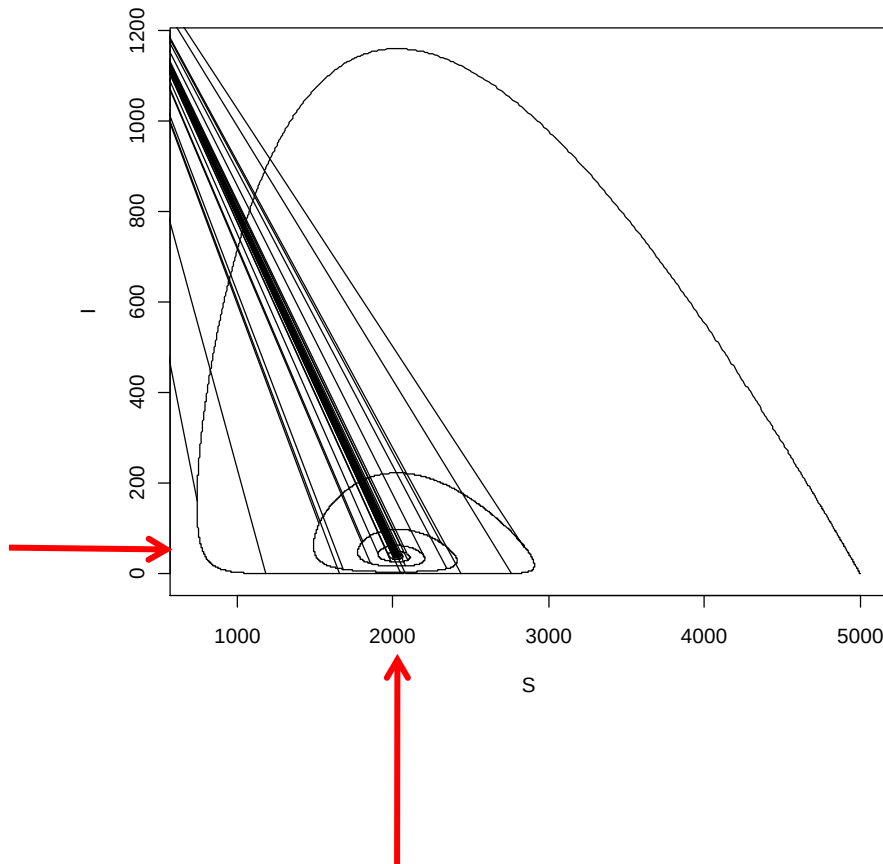
S



I



Equilibrium



At the long run, the infection reach the endemic equilibrium state in which at each time unit there are the same number of susceptible and infected individuals in the population.

Running the model in R

- To produce the model's output with **beta=0.001**:
 - Open R/R studio.
 - Open the program:
 - [ModelingIDinR1_V1_Stat&Dynam_10_2026.R](#).

Slides 52-65: the HTML file (example 1)

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er_prog1_STEAM3D_2026-V1.html Open in Browser Find Publish

- 1. The SIR model
- 2. The SIR model in a closed population
- 3. The SIR model in open population**
 - Dynamic aspects of the SIR model
 - Equilibrium values
 - Example 1 ($\beta=0.001$)
 - Example 2: changing the transmission rate ($\beta = 0.0005$)
 - Example 3: changing population size
- 4. Threshold and Equilibrium Values

3. The SIR model in open population

Dynamic aspects of the SIR model

We consider an open population model in which the birth rate is equal to the death rate, $\mu(a, t)$. We assume life long immunity and that there is no access mortality related to disease. The model formulated below follows individuals over time. This implies that the time unit of interest is the calendar time t .

$$\begin{aligned}\frac{\partial S(t)}{\partial t} &= N\mu(t) - (\lambda(t) + \mu(t))S(t), \\ \frac{\partial I(t)}{\partial t} &= \lambda(t)S(t) - (\nu + \mu(t))I(t), \\ \frac{\partial R(t)}{\partial t} &= \nu I(t) - \mu(a)R(t),\end{aligned}$$

The above model yields a constant population because of equal birth and death rates:

$$\frac{\partial S(t)}{\partial t} + \frac{\partial I(t)}{\partial t} + \frac{\partial R(t)}{\partial t} = \mu(N - S(t) - I(t) - R(t)) = 0.$$

Equilibrium values

The equilibrium values, $S_\infty, I_\infty, R_\infty$ are the number of individuals in each compartment when $t \rightarrow \infty$ and given by

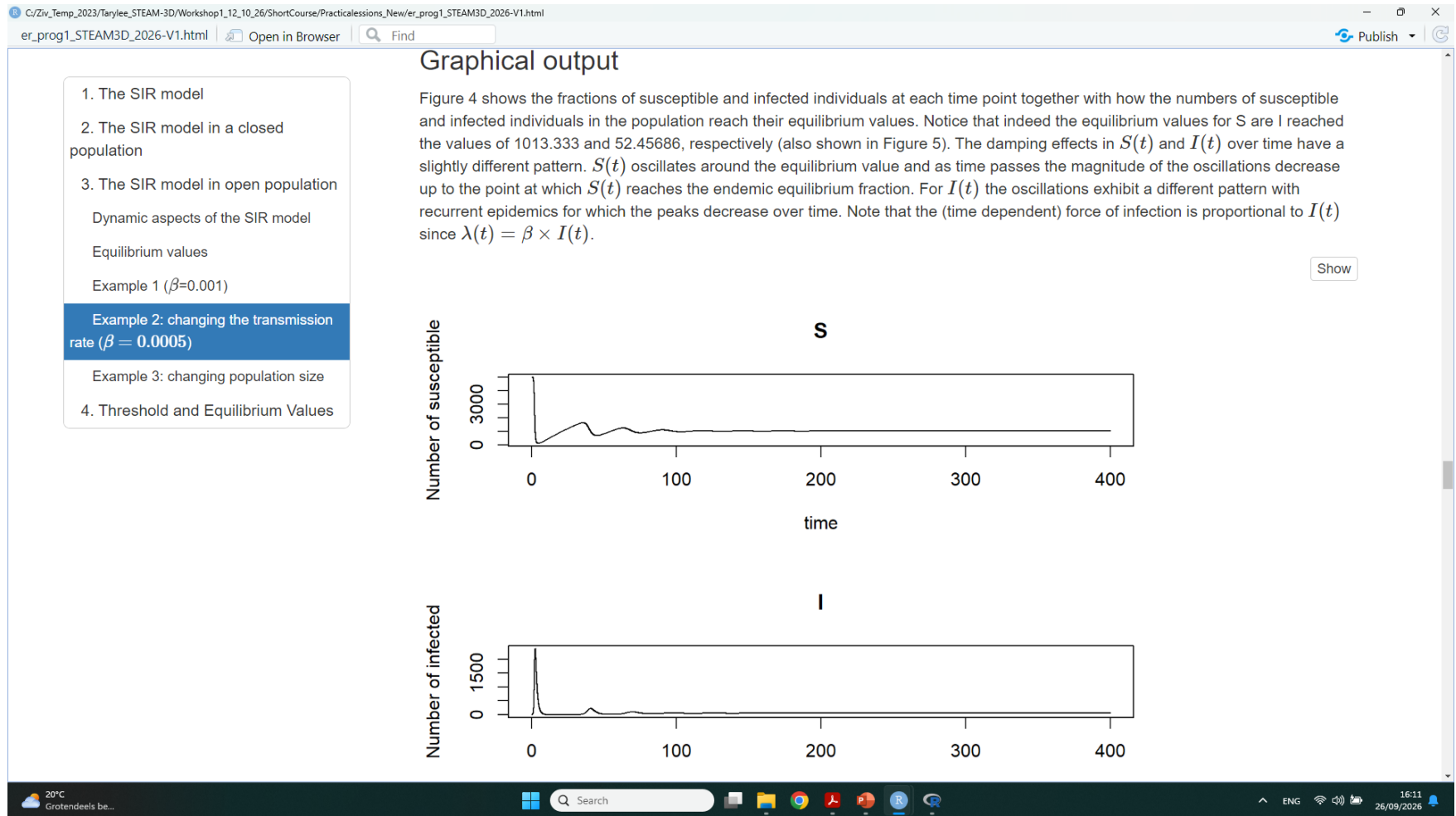
- $S_\infty = \frac{\nu + \mu}{\beta} \times N$.
- $I_\infty = \frac{\mu}{\beta} (R_0 - 1)$.
- $R_\infty = N - S_\infty - I_\infty$.

See Section 4 below for a numerical example.

Example 1 ($\beta=0.001$)

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The HTML file: model's solutions



Changing the parameters setting

- We consider two examples in which we change the parameters setting:

1. How the transmission changes when we change the transmission rate ?

$$\beta = 0.001 \rightarrow \beta = 0.0005$$

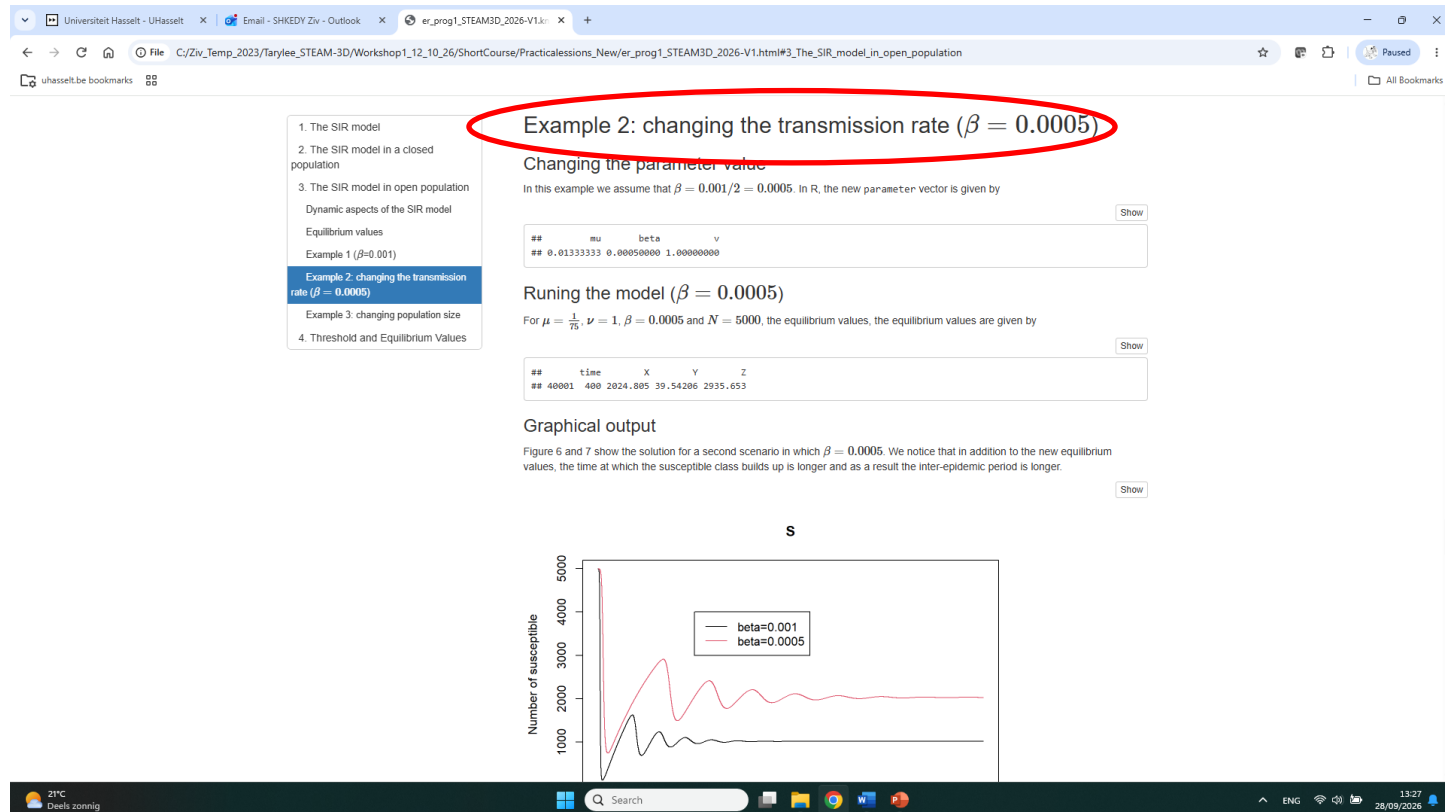
2. How the transmission changes when we change the population size ?

$$N = 5000 \rightarrow N = 2500$$

Changing the parameter setting (example 2)

- Changing the transmission rate:

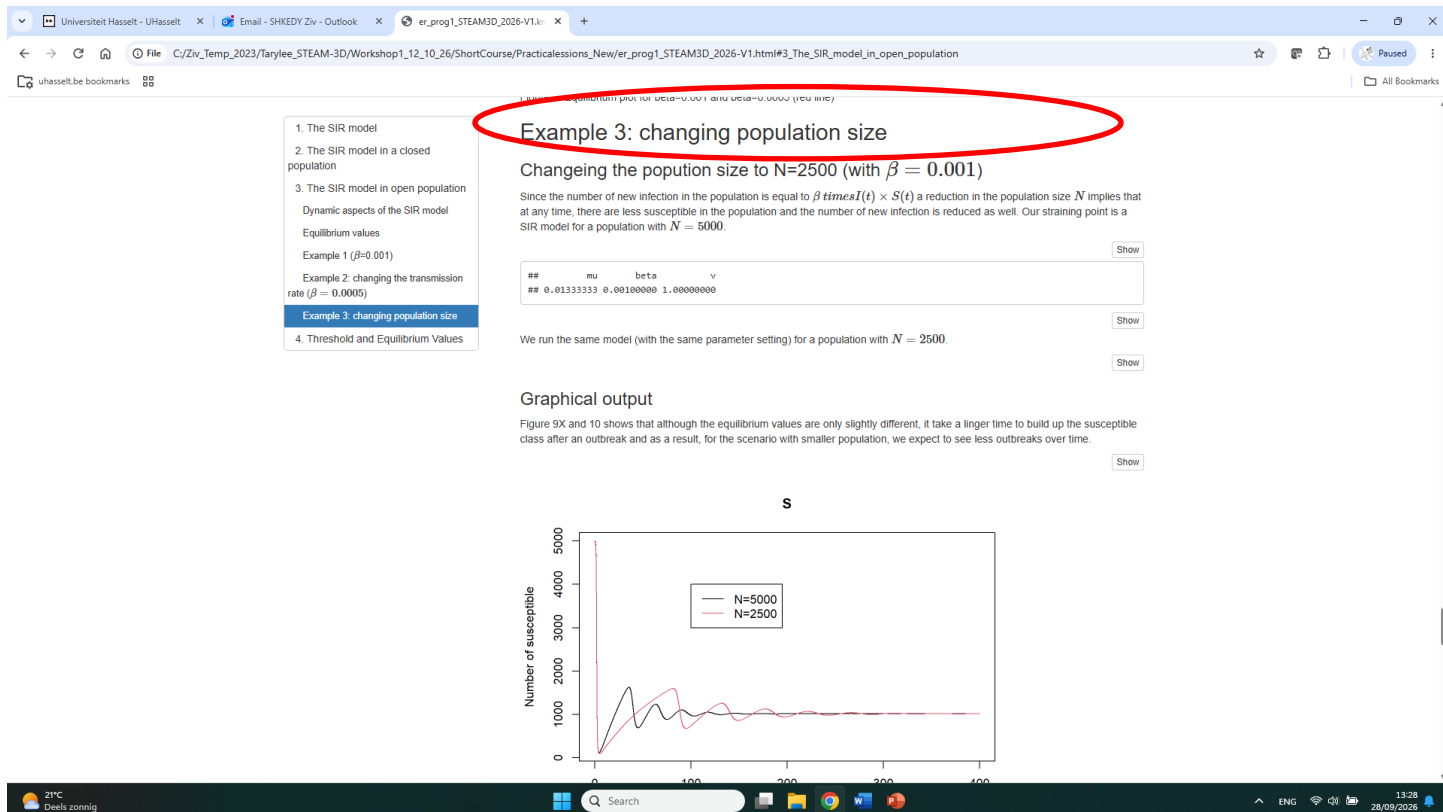
$$\beta = 0.001 \rightarrow \beta = 0.0005$$



Changing the parameter setting (example 3)

- Changing the population size:

$$N = 5000 \rightarrow N = 2500$$



Running the model in R

- Make the two examples by yourself.
- Look first at the code and output in the HTML.
- To produce the model's output:
 - Open R/R studio.
 - Open the program:
 - [ModelingIDinR1_V1_Stat&Dynam_10_2026.R.](#)

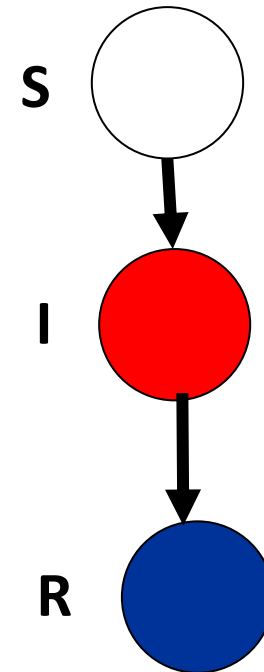
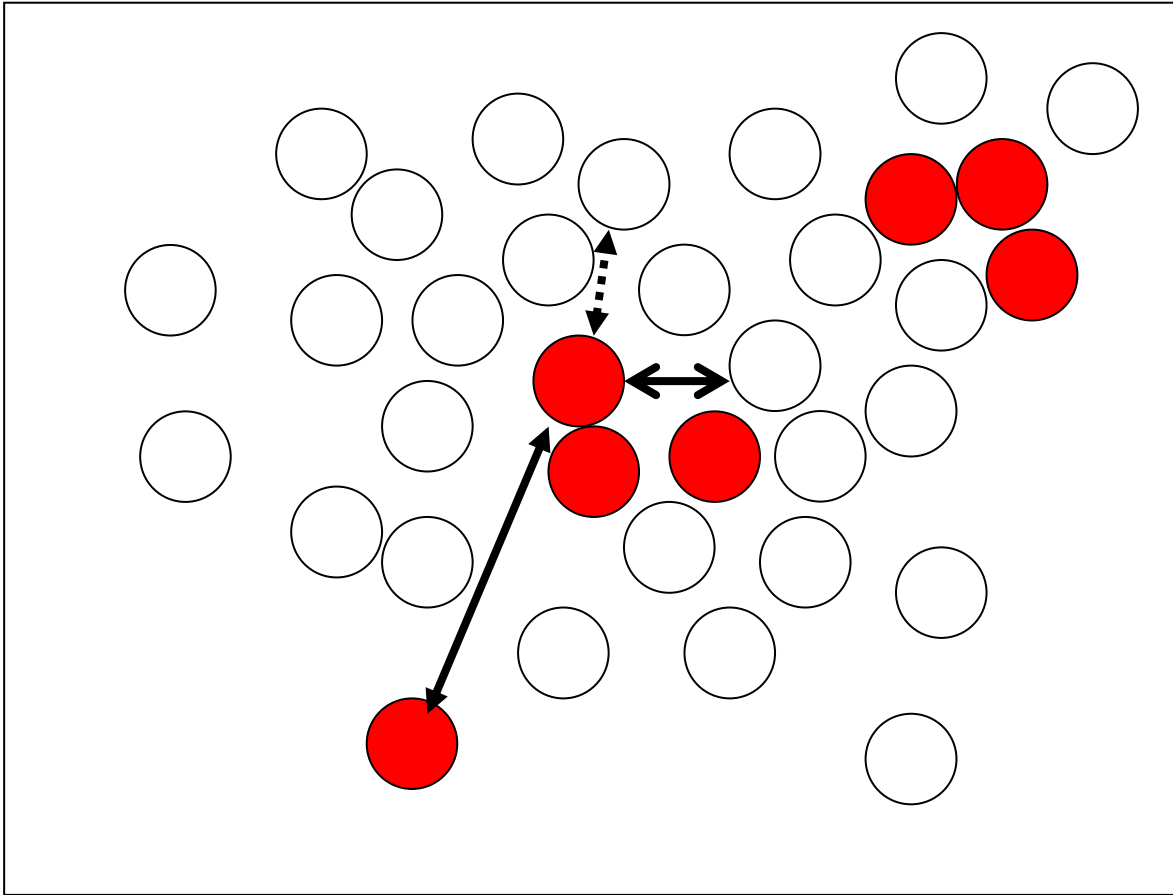
Part 4:

Vaccination in SIR model

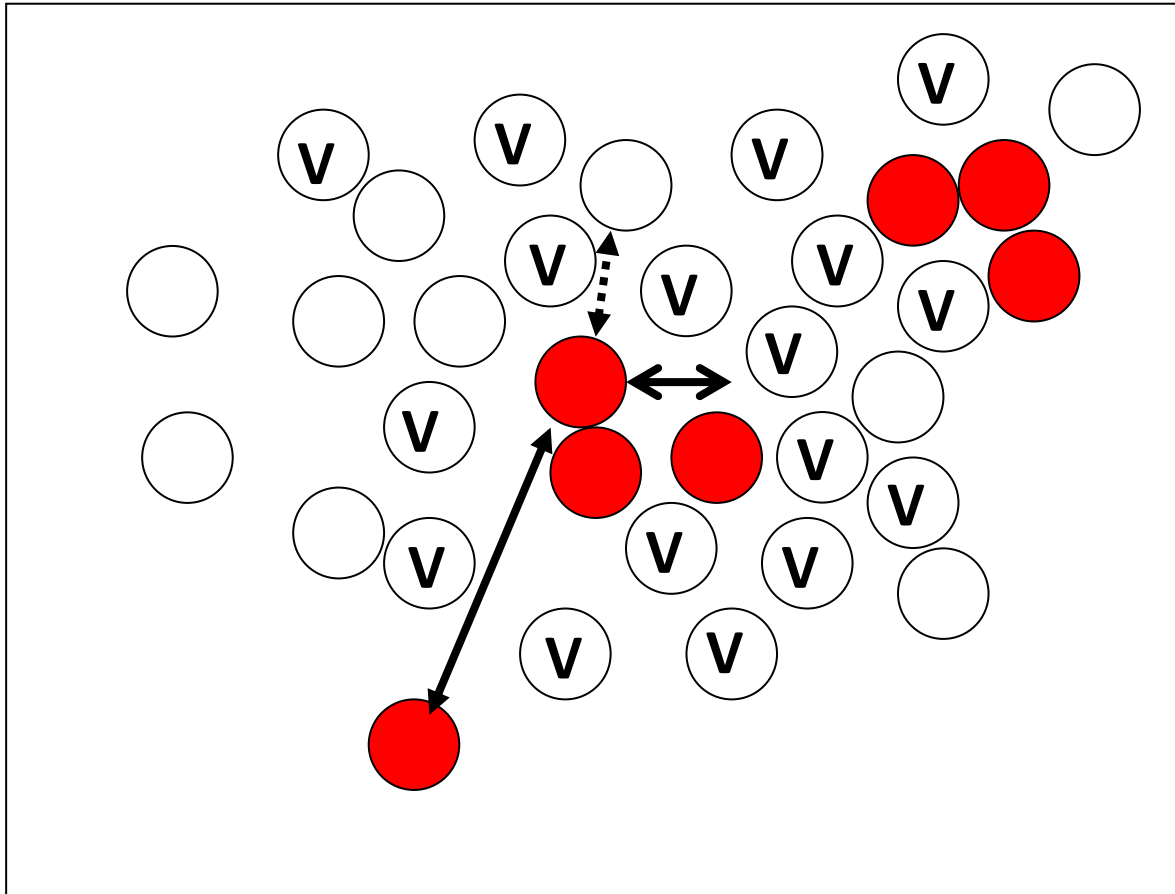
R programs for the practical session

- HTML file for slides 81-99:
 - [eR_prog2_STEAM3D_2026_V1.html](#)
- Two R programs:
 - R program:
 - [ModelingIDinR4_V1_Vaccination_10_2026.R](#).
 - R requirements: basic R editor.
 - R Markdown:
 - [eR_prog2_STEAM3D_2026_V1.Rmd](#).
 - R requirements: R Studio.
 - This program produces the HTML file

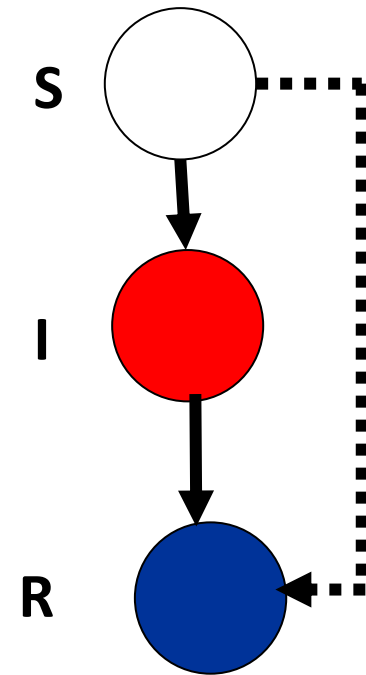
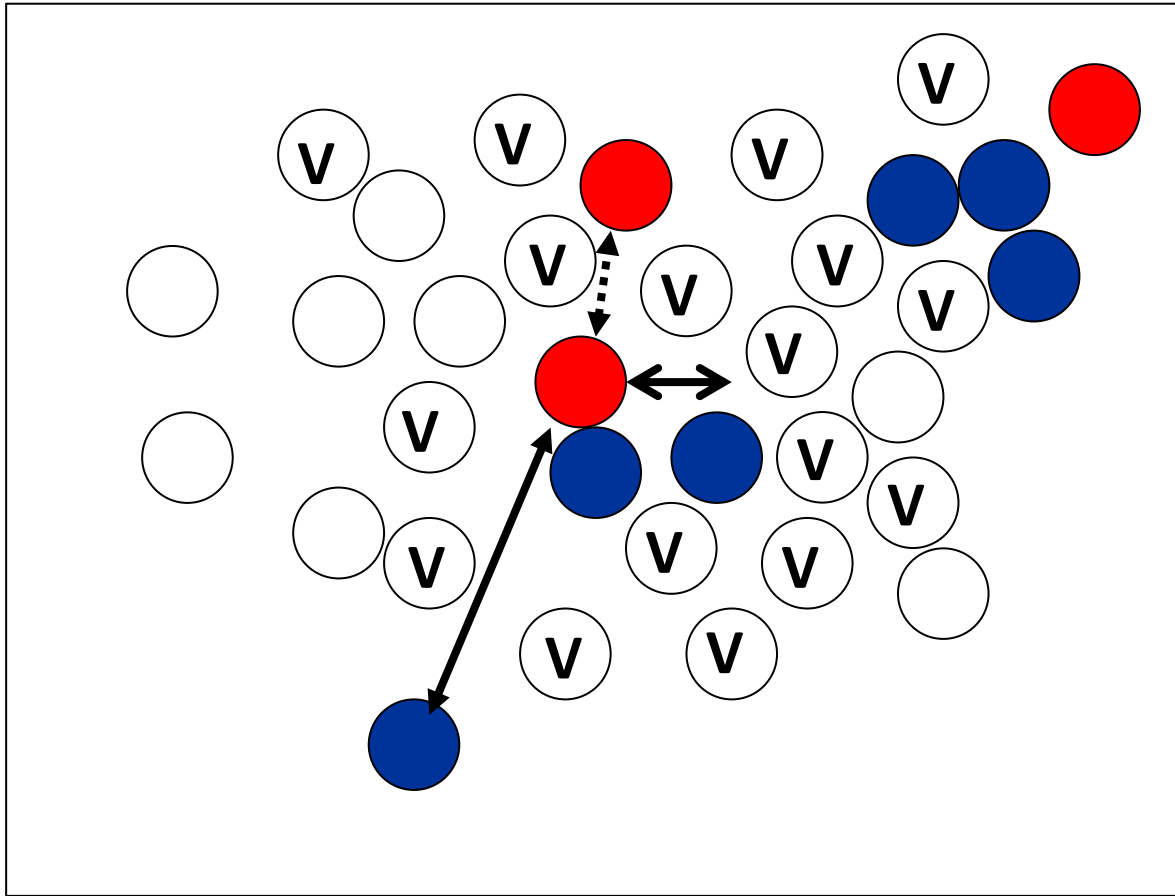
Vaccination



Vaccination

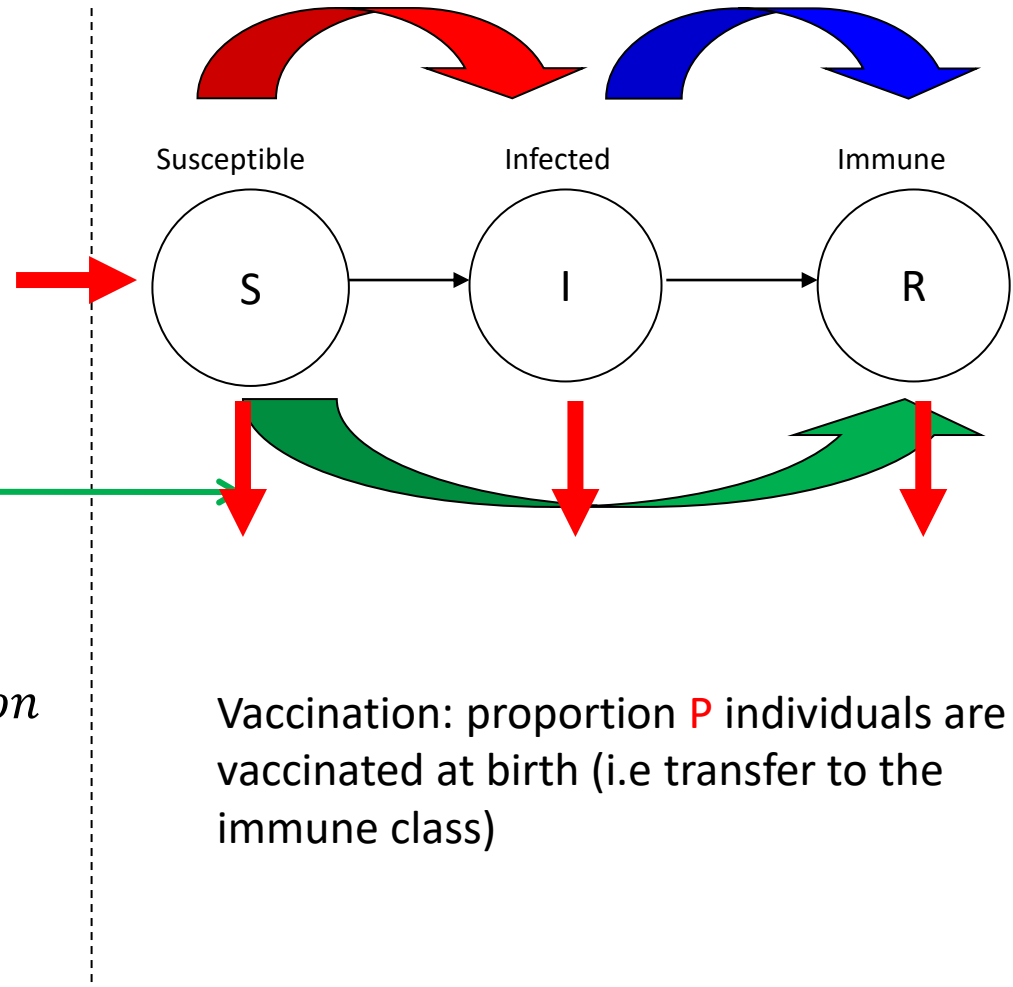


Vaccination



Transmission model with vaccination

$$\begin{aligned}\frac{dS(t)}{dt} &= N\mu \times (1-P) - \lambda S - \mu S \\ \frac{dI(t)}{dt} &= \lambda S - (\sigma + \mu)I \\ \frac{dR(t)}{dt} &= \sigma I - \mu R + N\mu \times P\end{aligned}$$



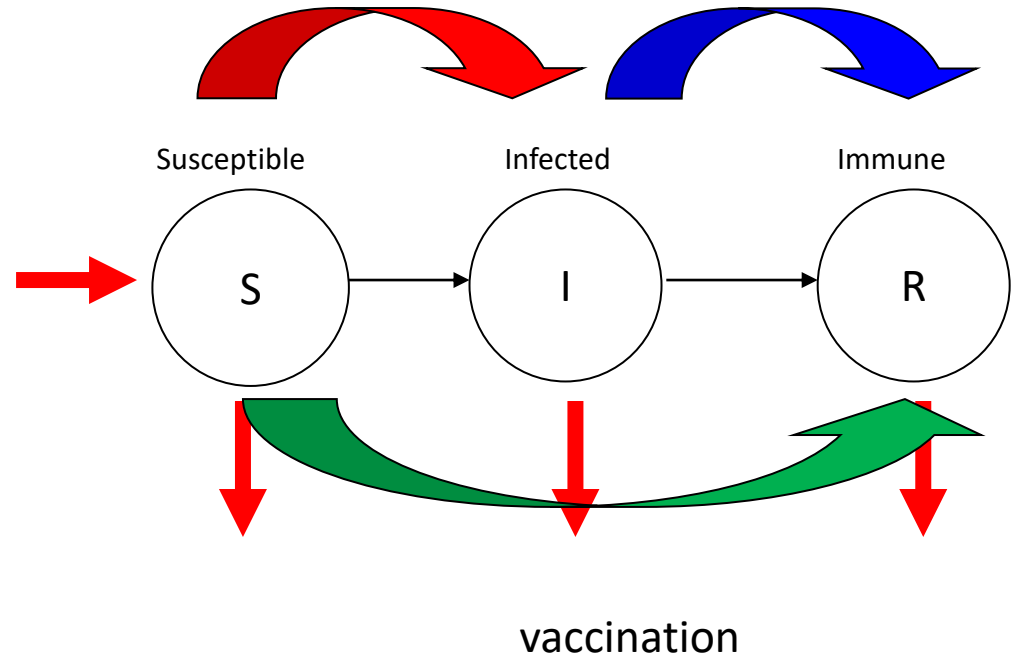
$\lambda(t) = \beta \times I(t)$: force of infection

Vaccination: proportion **P** individuals are vaccinated at birth (i.e transfer to the immune class)

SIR transmission Model with vaccination

Model parameters:

1. Life expectancy: 75 years.
2. Transmission rate 0.0005.
3. Recovery rate 1 time unit.
4. Vaccination **45%** at birth.

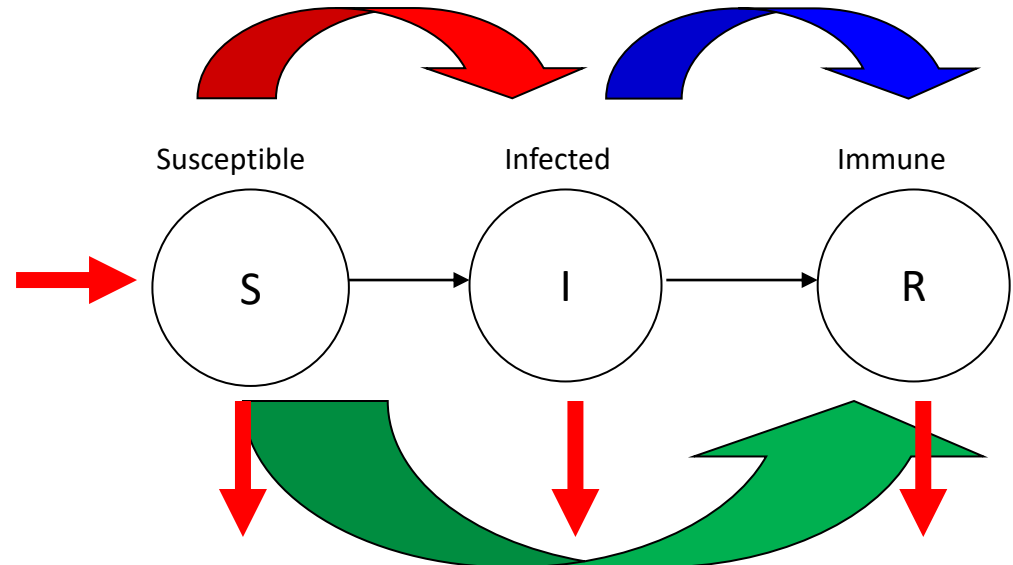


```
> parameters <- c(mu=1/75,beta=0.001/2, v=1, P=0.45)
> parameters
      mu      beta      v      P
0.01333333 0.00050000 1.00000000 0.45000000
```

SIR transmission model with vaccination

- The model in R:

```
SIR<-function(t,state,parameters)
{
  with(as.list(c(state, parameters)),
  {
    dX <- 5000*mu*(1-P)-beta*Y*X - mu*X
    dY <- beta*Y*X - v*Y - mu*Y
    dZ <- v*Y -mu*Z+5000*mu*P
    list(c(dX, dY, dZ))
  })
}
```



β = transmission rate.
 N = population size.
 μ = birth rate = death rate.
 v = recovery rate.
 p = vaccinate rate.

$$\left\{ \begin{array}{l} \frac{dS(t)}{dt} = N\mu \times (1 - P) - \beta IS - \mu S \\ \frac{dI(t)}{dt} = \beta IS - (v + \mu)I \\ \frac{dR(t)}{dt} = vI - \mu R + N\mu \times P \end{array} \right.$$

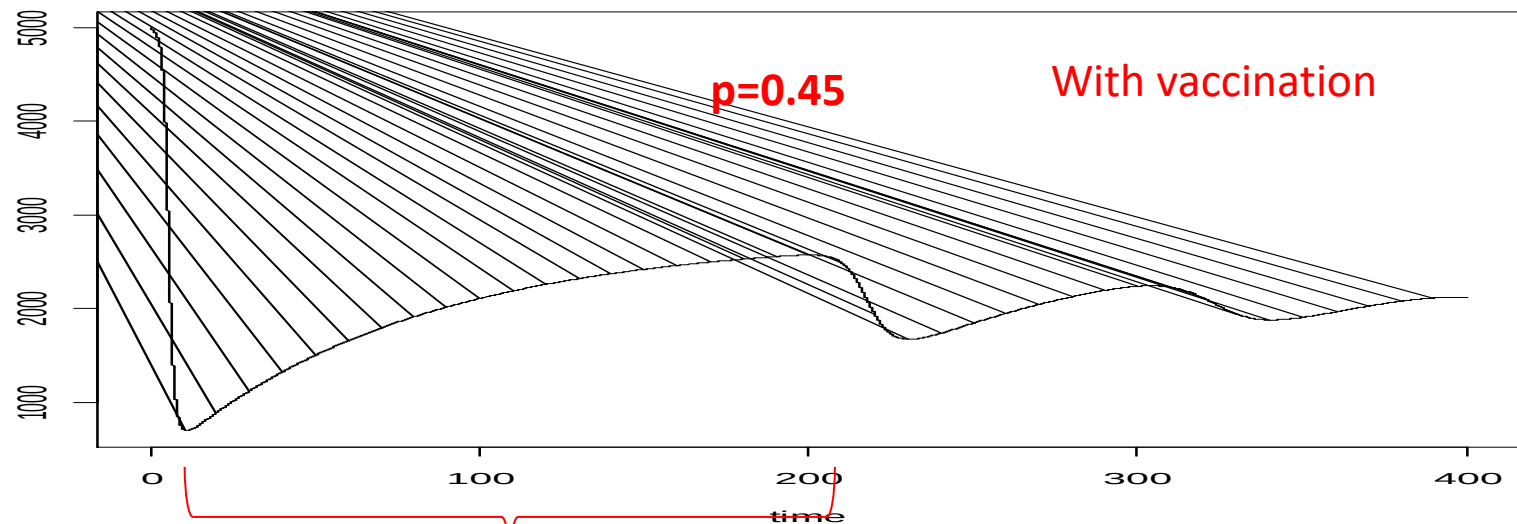
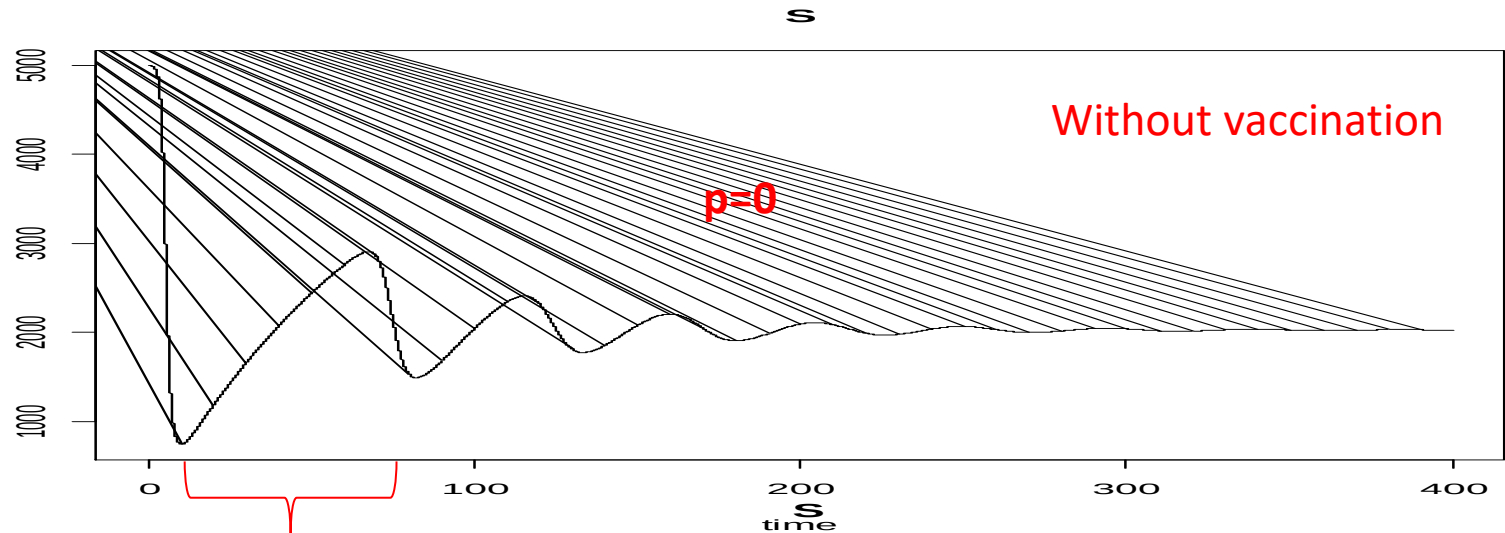
Part 4.1:
Example 1: SIR model with $p=0.45$

SIR model with vaccination at birth

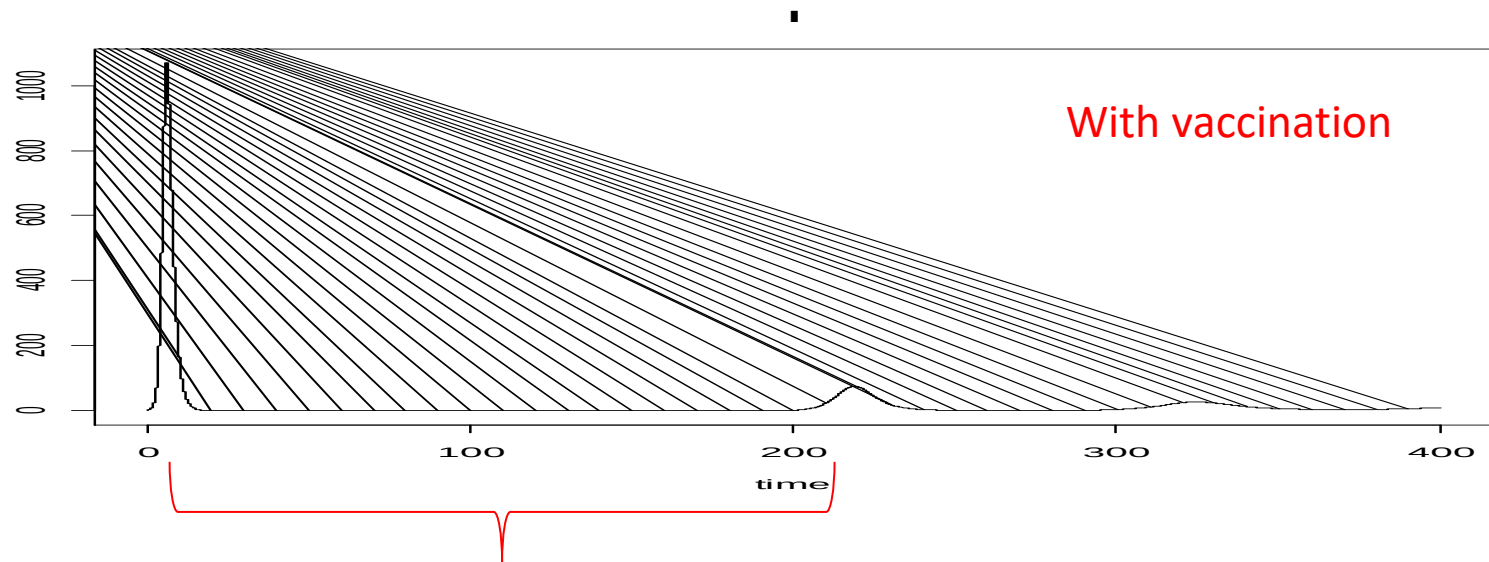
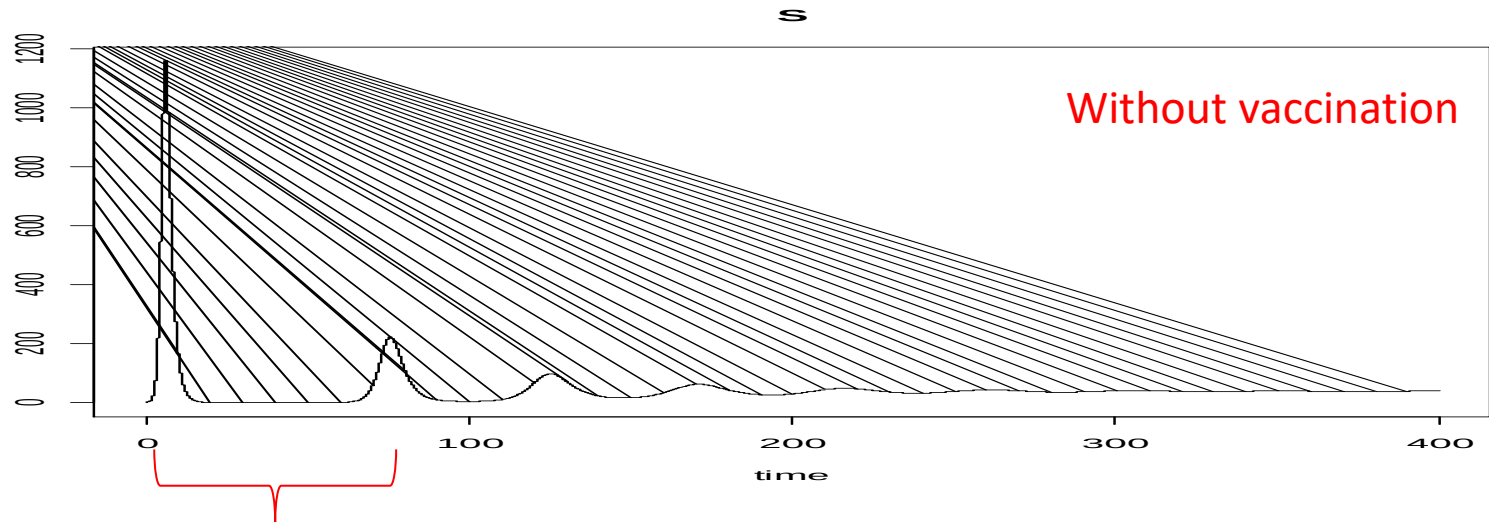
45% are vaccinated at birth.

```
> parameters <- c(mu=1/75,beta=0.001/2, v=1, P=0.45)  
> parameters  
           mu           beta           v           P  
0.01333333 0.00050000 1.00000000 0.45000000
```

Susceptible: inter epidemic period

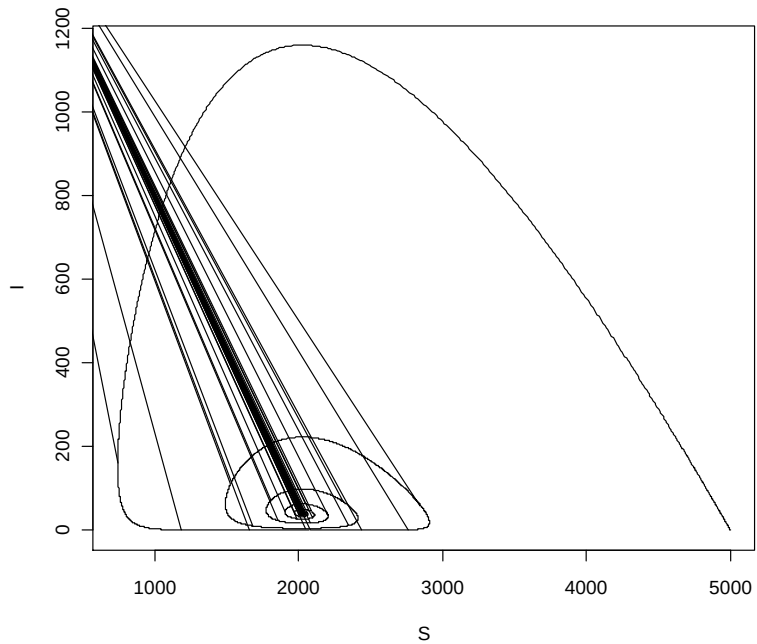


Infected class: inter epidemic period

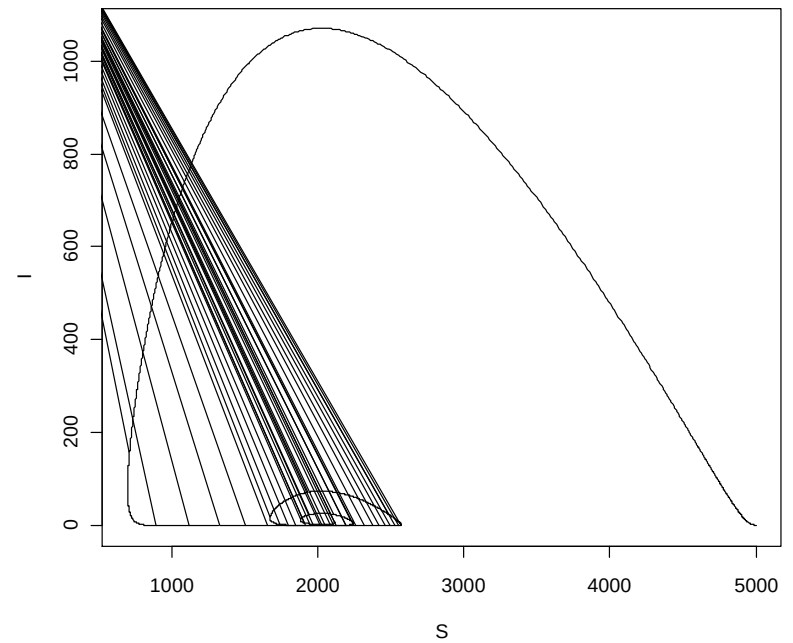


Equilibrium

Without vaccination



With vaccination



Part 4.2:

Example 1: SIR model with $p=0$ and $p=0.4$

Parameter setting

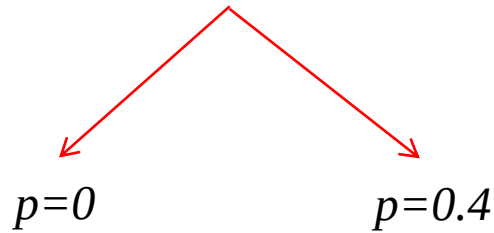
Two scenarios:

$$\beta = 0.0005.$$

$$N=5000.$$

$$\mu = \frac{1}{75}.$$

$$v = 1.$$



Parameter setting in R

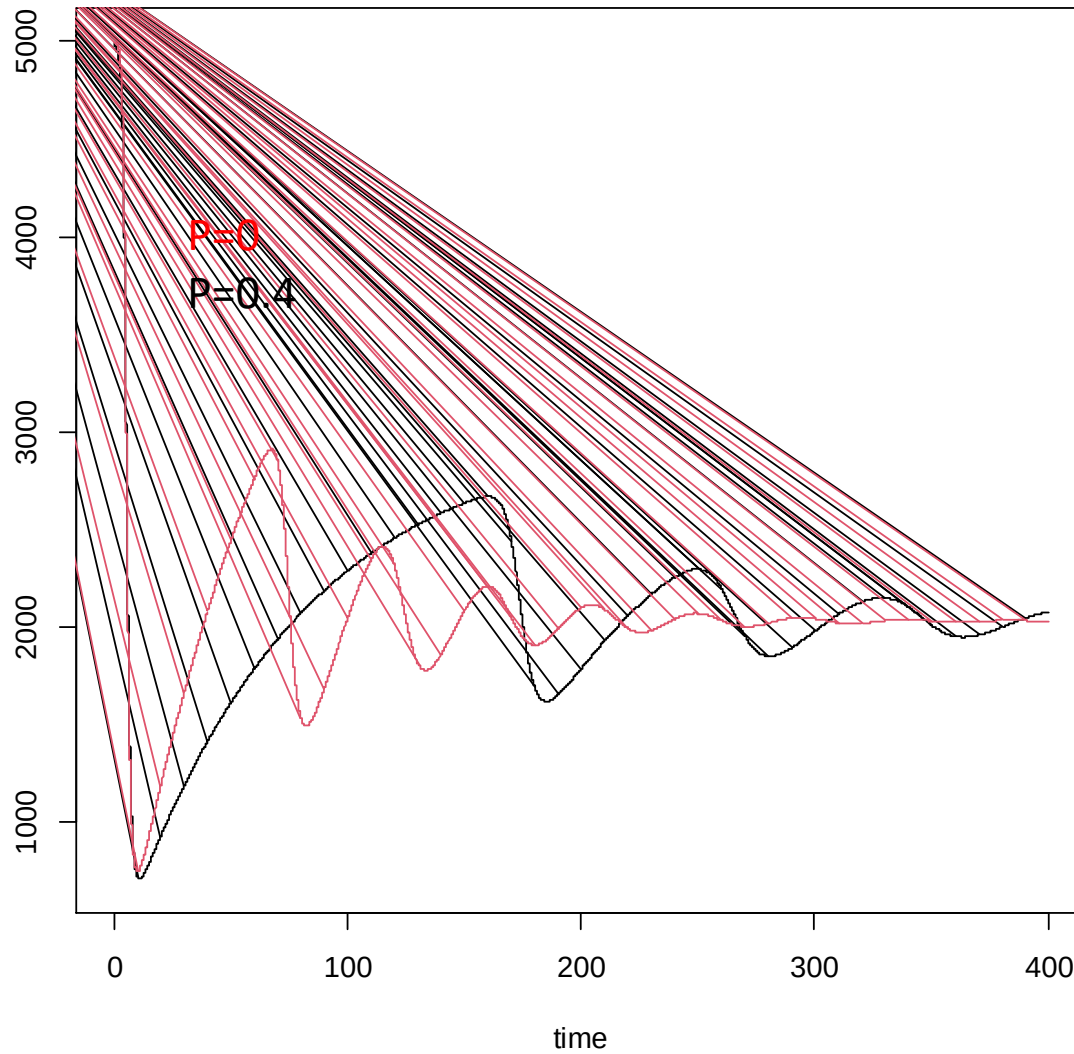
```
> stateparameters <- c(mu=1/75,beta=0.001/2, v=1, P=0.40)
> parameters
      mu      beta      v      P
0.01333333 0.00050000 1.00000000 0.40000000
> state <- c(X=4999,Y=1,Z=0)
> state
   X    Y    Z
4999  1    0
>
```



$S(0) = 4999, I(0) = 1, R(0) = 0$

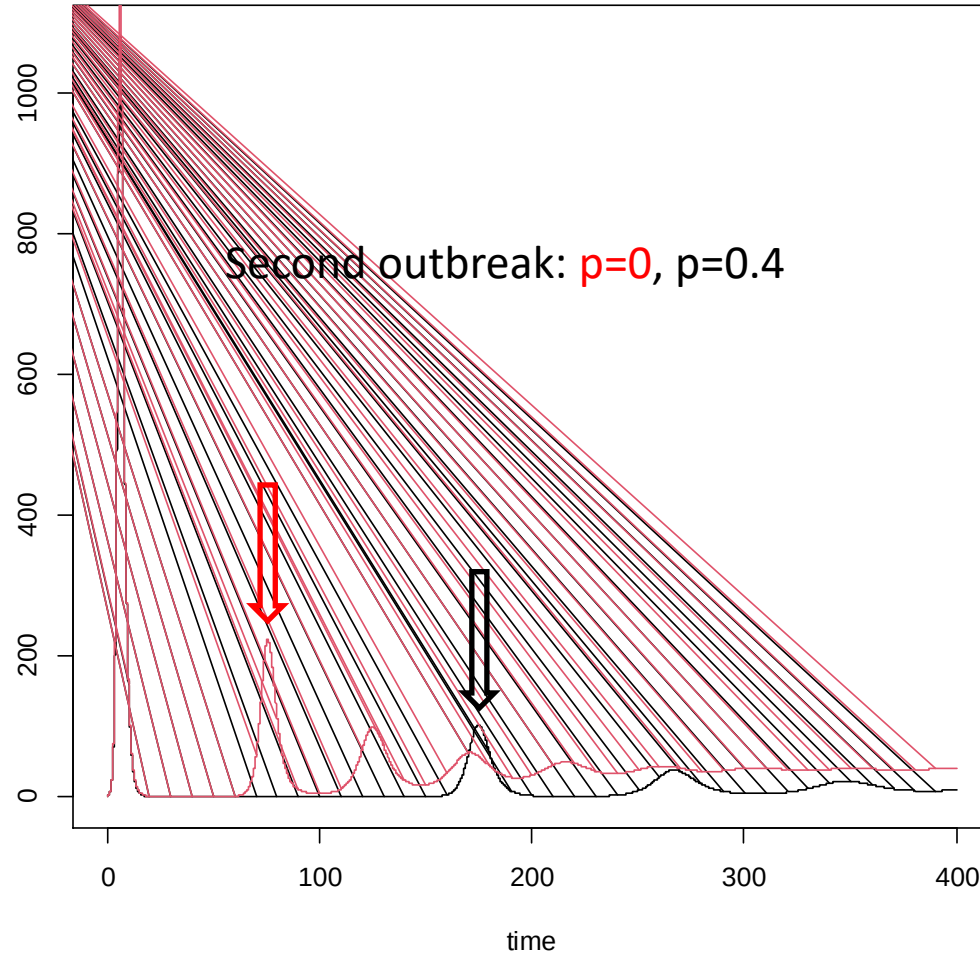
Susceptible

s

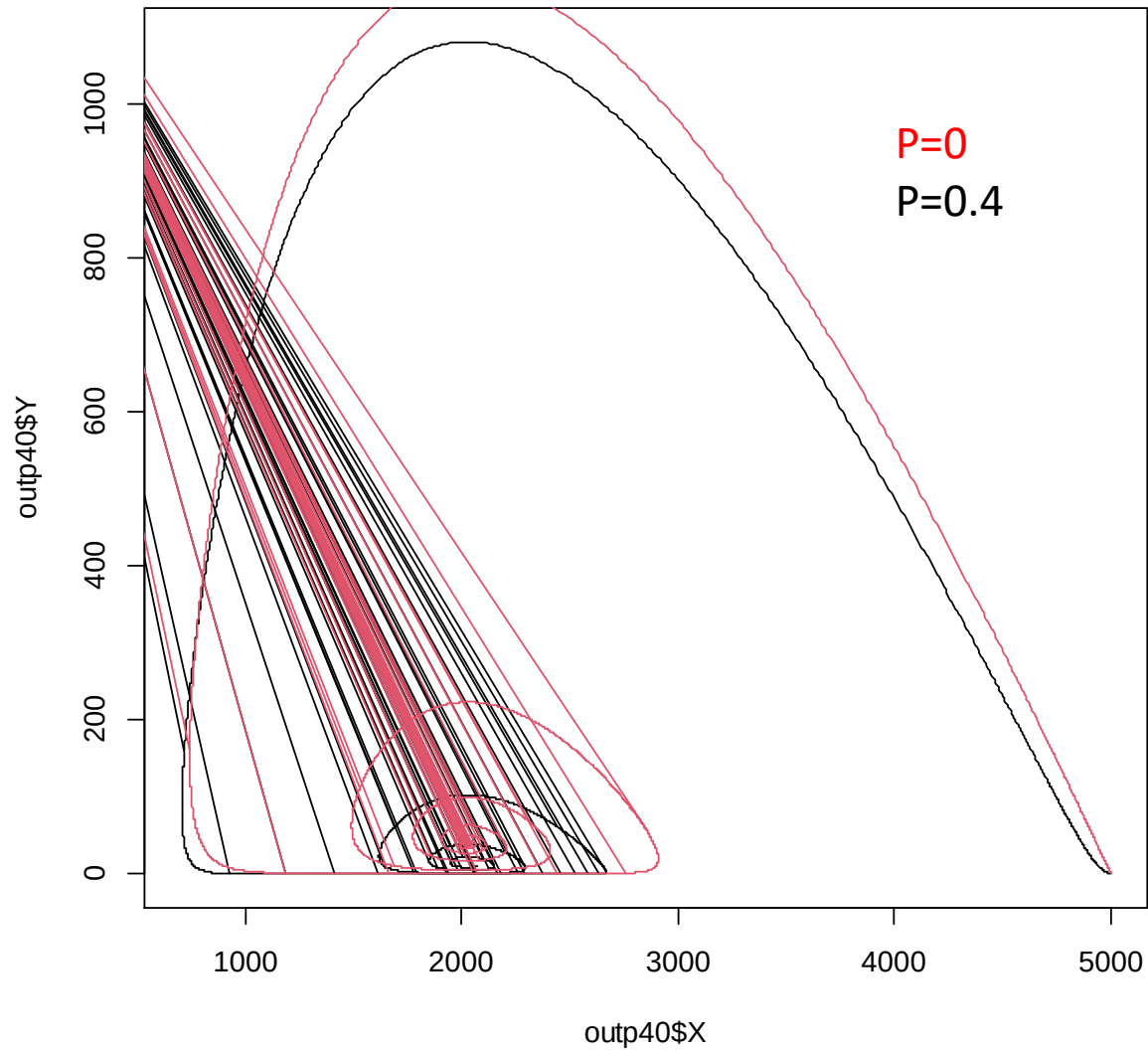


Infected

I



Equilibrium values



Slides 92-97: the HTML file

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Code

24-09-2026 >eR-BioStat and STAEM-3D

Modeling infectious diseases using R - vaccination in the SIR model

Ziv Shkedy and Rashider Aloni based on Chapter 2-4 in the book *Modeling infectious diseases parameters based on serological and social contact data* (<https://www.simid.be/modeling-infectious-disease-parameters-based-on-serological-and-social-contact-data/>)

Show

1. The SIR model with vaccination in open population

Incorporating a vaccination program in the SIR model can be done by adding a direct path from the susceptible compartment to the immune compartment. We illustrate this using a SIR model that is age-independent and we look at its dynamic behavior over time. Let us assume that a proportion p of each cohort of newborns is vaccinated at birth, the SIR model with demography can then be written as

$$\begin{cases} \frac{dS}{dt} = N\mu(1-p) - (\lambda(t) + \mu)S, \\ \frac{dI}{dt} = (\lambda(t) + \mu)S - (\nu + \alpha + \mu)I, \\ \frac{dR}{dt} = N\mu p + \nu I - \mu R. \end{cases}$$

The SIR model formulated above implies that, for each cohort, a proportion p enter the population directly in the immune class and therefore do not take a part in the transmission process.

2. The SIR model with vaccination in R: post vaccination equilibrium

Recall that the basic reproductive number, R_0 , is defined as

$$R_0 = \frac{N \times \beta}{\nu + \mu}.$$

If R_0 is above the threshold of one, the disease will maintain itself while for $R_0 < 1$ the disease will die out.

The Critical Proportion of Vaccination

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Slides 92-97: the HTML file - example 2

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1. The SIR model with vaccination in open population

2. The SIR model with vaccination in R: post vaccination equilibrium

3. Example 1: a model without vaccination ($p = 0$)

4. Example 2: model with 40% vaccination at birth

Definition of the model parameters

Model solution

5. Example 3: The critical proportion of vaccination

4. Example 2: model with 40% vaccination at birth

Definition of the model parameters

In this example we assume that 40% of the population are vaccinated at birth. The state variables remain the same as before

Show

```
##      X      Y      Z
## 4999      1      0
```

but we need to specify the proportion of vaccination. For a population size of 5000, $\beta = 0.0005$, $\nu = 1$, life expectancy of 75 years and vaccination rate of 40% at birth, $p = 0.4$ we have

Show

```
##      mu      beta      v      P
## 0.0133333 0.0005000 1.0000000 0.4000000
```

Model solution

Figure 3 shows the number of susceptible over time for two scenarios with $p = 0$ and $p = 0.4$ and reveals the post vaccination equilibrium patterns. The inter epidemic period is longer when the proportion of vaccinated individuals increases. This implies that individuals stay, on average, longer time in the susceptible class and as a result the average age in which individuals are infected (the average age at infection) increases. In the long run, after vaccination, the population will research a new endemic equilibrium. Notice that although the pre-vaccination force of infection $\lambda(0)$ is equal under the two scenarios, in the new equilibrium the force of infection decreases as the proportion of vaccinated individuals increases. Since $\lambda(t)$ decreases as p increases the rate in which individuals leave the susceptible class decreases as p increases and as a result, as shown in Figure 4 the inter epidemic period is longer as p increases. This implies that a vaccination program has two effects on the population:

- The direct effect is the transfer of a proportion p from the susceptible directly to the immune class.
- The indirect effect is related to the decline of the magnitude of the force of infection and its affect both vaccinated and unvaccinated individuals.

Show

```
##      time      X      Y      Z
## 1 0.00 4999.000 1.000000 0.000000
## 2 0.01 4998.708 1.014972 0.2767231
## 3 0.02 4998.416 1.030167 0.5535601
## 4 0.03 4998.124 1.045588 0.8305133
## 5 0.04 4997.831 1.061237 1.1075846
## 6 0.05 4997.538 1.077120 1.3847767
```

Show

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Running the model in R

- Make the two examples (with $p=0$ & $p=0.4$) by yourself.
- Look first at the code and output in the HTML file.
- To produce the model's output:
 - Open R/R studio.
 - Open the program:
 - [ModelingIDinR4_V1_Vaccination_10_2026.R](#).